Neutrinos: Masses, Mixings, and Detection

A Review of the Phenomenology

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Outline

1. Introduction and Early Phenomenology
2. Atmospheric Neutrino Oscillations
3. Neutrino Oscillations in Matter (MSW Effect)
4. The Solar Neutrino Problem
5. Three-Neutrino Framework
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The Neutrino Hypothesis - Pauli's Proposal (1930)

- In a **two-body decay**, the energy of daughter particles is *fixed* by kinematics.
- In β-decay $n \to p + e^-$, the electron energy spectrum is **continuous**, not monoenergetic.
- Two explanations:
  1. Energy conservation is violated (Bohr's suggestion – rejected).
  2. A third, neutral, nearly massless particle carries away the missing energy.
- Pauli postulated a new neutral particle (originally called "neutron"); Fermi renamed it **neutrino** (ν) in 1934.
- Fermi then built the first quantitative theory of β-decay using this hypothesis.

Beta-decay spectrum

$$\frac{d\Gamma}{dE} \propto p_e E_e (E_0 - E_e)\sqrt{(E_0 - E_e)^2 - m_\nu^2}$$

$E_0 = Q - m_e - m_\nu$: endpoint energy

The *shape* near the endpoint directly constrains m_ν.

Direct Mass Measurements from Beta Decay

- The endpoint of the β-spectrum depends sensitively on m_{ν_e}.
- The **KATRIN** experiment (Karlsruhe, using tritium) currently gives [1]: $m_{\nu_e} < 0.8\ \mathrm{eV} \quad (90\%\ \mathrm{CL})$
- The mass of ν_μ is constrained from the monoenergetic muon in $\pi^+ \to \mu^+ + \nu_\mu$ (two-body decay at rest): $m_{\nu_\mu} < 100\ \mathrm{keV}$
- The mass of ν_τ is bounded by τ decays: $m_{\nu_\tau} < \ \mathrm{MeV}$.[1]

Important caveat

These limits are in a *pre-neutrino-mixing* spirit. Since the mass eigenstates differ from flavour eigenstates, and $|\Delta m^2| < 10^{-2}\ \mathrm{eV}^2$, the keV/MeV limits are no longer phenomenologically relevant.

[1] Source: Notes by Amol Dighe, TIFR[2]

Discovery of the Three Neutrino Flavours

Electron neutrino ν_e - Reines & Cowan (1956)

Directed $\bar{\nu}_e$ from a reactor into a water + $CdCl_2$ target:

$$\bar{\nu}_e + p \rightarrow n + e^+$$

Positrons annihilated giving two 0.5 MeV photons; neutrons captured by Cd emitting γ's within a few μs. Coincidence of both signals $\Rightarrow$ confirmed ν_e.

Muon neutrino ν_μ - Lederman, Schwartz, Steinberger (1962)

AGS proton beam $\rightarrow$ pions $\rightarrow \nu_\mu$; detected $\nu_\mu + N \rightarrow \mu + N'$. Only muons appeared, *never* electrons $\Rightarrow \nu_\mu \neq \nu_e$. *(Nobel Prize 1988)*

Tau neutrino ν_τ - DONuT experiment, FNAL (2000) [3]

High-energy proton beam $\rightarrow D_s \rightarrow \tau \rightarrow \nu_\tau$; detected $\nu_\tau + N \rightarrow \tau + N'$. Difficulty: τ lifetime $\approx 2.9 \times 10^{-13}$ s, very short tracks.

Parity Violation and the V−A Theory

- **Wu et al. (1956):** ^{60}Co polarised by a magnetic field at low T; electrons emitted *preferentially opposite* to the nuclear spin direction.
- **Telegdi (1957):** Angular distribution of electrons from polarised muon decay also showed parity violation.
- Both experiments violated parity *maximally*: the weak interaction couples **only to left-handed particles and right-handed antiparticles**.

Fermi's 4-Fermi interaction (beta decay)

$$\mathcal{L} = \frac{G_F}{\sqrt{2}}\left[\bar{p}\,\gamma^\mu(C_V - C_A\gamma^5)n\right]\left[\bar{e}\,\gamma_\mu(1-\gamma^5)\nu_e\right]$$

The $(1-\gamma^5)$ projects onto the *left-handed* component only. This is a low-energy effective description; the actual mediator is the W boson.

Neutrino Interactions in the Standard Model

Charged Current (CC) - mediated by $W^{\pm}$:

$$\mathcal{L}_{CC} = -\frac{g}{2\sqrt{2}}\left(\bar{\ell}\,\gamma^{\mu}(1-\gamma^{5})\nu_{\ell}\,W_{\mu}^{-} + \text{h.c.}\right)$$

Only left-handed neutrinos participate; W couples equally to all flavours.

Neutral Current (NC) - mediated by Z^{0}:

$$\mathcal{L}_{NC} = -\frac{g}{2\cos\theta_{W}}\left(\bar{f}\,\gamma^{\mu}(C_{V}^{f} - C_{A}^{f}\gamma^{5})f\,Z_{\mu} + \text{h.c.}\right)$$

For left-handed neutrinos ($T_{3}^{\nu} = +\frac{1}{2}$, $Q_{\nu} = 0$):

$$C_{V}^{\nu} = C_{A}^{\nu} = \frac{1}{2}$$

Right-handed neutrinos have $T_{3}^{\nu} = 0$, $Q_{\nu} = 0 \Rightarrow$ **no SM interaction**.

Number of Neutrino Species - Z Boson Decay Width

- Total decay width of the Z boson: $\Gamma_Z = 2495$ MeV [4].
- Visible contributions (quarks + charged leptons): $\Gamma_{\text{vis}} \approx 1996$ MeV.
- **Invisible width:** $\Gamma_{\text{inv}} = \Gamma_Z - \Gamma_{\text{vis}} \approx 499$ MeV.
- Each active neutrino species contributes ≈ 167 MeV:

$$N_\nu = \frac{499}{167} \approx 3.0$$

- Conclusively: **exactly 3 light active neutrinos**.

Caveats

Additional neutrino species are allowed *only if*:

- They are **sterile** (no SM gauge coupling), or
- They are heavier than $M_Z/2 \approx 45.6$ GeV.

Atmospheric Neutrinos - Production and Expected Flux

When cosmic rays strike the atmosphere:

$$\pi^{\pm} \to \mu^{\pm} + \nu_{\mu}\,(\bar{\nu}_{\mu})$$
$$K^{\pm} \to \mu^{\pm} + \nu_{\mu}\,(\bar{\nu}_{\mu})$$
$$\mu^{\pm} \to e^{\pm} + \nu_{e}\,(\bar{\nu}_{e}) + \bar{\nu}_{e}\,(\nu_{\mu})$$

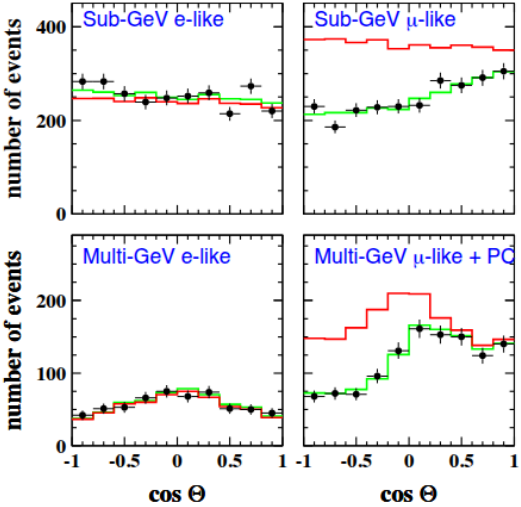
Expected flavour ratio (ignoring $\nu/\bar{\nu}$):

$$\Phi(\nu_{\mu}) : \Phi(\nu_{e}) \approx 2 : 1$$

What was observed (SK[5])

- ν_e flux: **consistent** with expectation.
- ν_μ flux: **significantly deficit**.
- The deficit was *zenith-angle dependent*.

Atmospheric Neutrino Flux vs. Zenith Angle



Red: no-oscillation expectation. Green+dots: observed flux. Top: sub-GeV; bottom: multi-GeV. Left: ν_e; right: ν_μ.

Super-Kamiokande Observations (1998)

Key features of the Super-Kamiokande data:

- Under no-oscillation hypothesis, the flux vs. $\cos\theta_z$ should be *symmetric* (modulo Earth's magnetic field effects, which are well understood).
- ν_e flux matched expectations at *all* energies and zenith angles.
- ν_μ flux was always *below* expectation for sub-GeV neutrinos.
- For multi-GeV neutrinos: ν_μ deficit appeared only for $\cos\theta_z < 0$ (upward-going, $L \sim$ diameter of Earth $\approx$ 13,000 km), but not for downward-going ($L \sim 15$ km).
- Smoking-gun: a dip in the L/E spectrum (shown later) confirmed oscillatory behaviour rather than decay.

Interpretation

The deficit depends on L/E and does not affect ν_e - consistent only with **neutrino oscillations** $\nu_\mu \to \nu_\tau$.

Two-Neutrino Oscillations in Vacuum - Formalism

Consider a two-flavour system (ν_μ, ν_τ). Flavour and mass eigenstates are related by:

$$\begin{pmatrix} \nu_\mu \\ \nu_\tau \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \end{pmatrix}$$

Ultra-relativistic approximation: $E_i \approx p + \dfrac{m_i^2}{2E}$, so effective energy $\epsilon_i = m_i^2/2E$.

A neutrino produced as $|\nu_\alpha\rangle$ at $t = 0$ evolves to:

$$|\nu(t)\rangle = \cos\theta\, e^{-iE_1 t}|\nu_1\rangle + \sin\theta\, e^{-iE_2 t}|\nu_2\rangle$$

Transition probability:

$$\boxed{P_{\alpha\to\beta} = \sin^2(2\theta)\,\sin^2\left(\frac{\Delta m^2\, L}{4E}\right)}$$

Survival probability:

$$P_{\alpha\to\alpha} = 1 - \sin^2(2\theta)\,\sin^2\left(\frac{\Delta m^2\, L}{4E}\right)$$

In experimental units: $P_{\alpha\to\beta} = \sin^2(2\theta)\,\sin^2\left(1.27\,\dfrac{\Delta m^2[\mathrm{eV}^2]\, L[\mathrm{km}]}{E[\mathrm{GeV}]}\right)$

Explaining Atmospheric Observations with Oscillations

The oscillation length is $\lambda \sim \dfrac{E}{\Delta m^2}$.

- **Sub-GeV, downward-going:** Short baseline $L \sim 15$ km, $\lambda \ll L \Rightarrow$ averaging over rapid oscillations gives a deficit $\sim \sin^2(2\theta)/2$.
- **Multi-GeV, downward-going:** $\lambda \gg L_{\rm atm} \Rightarrow \sin^2(\Delta m^2 L/4E) \ll 1$, negligible deficit. Consistent with observations.
- **Multi-GeV, upward-going:** $L \sim 10^4$ km, $\Delta m^2 L/E \gg 1 \Rightarrow$ significant deficit. Consistent with observations.
- ν_e unaffected $\Rightarrow$ oscillation is $\nu_\mu \leftrightarrow \nu_\tau$ (not $\nu_\mu \leftrightarrow \nu_e$).

Best-fit parameters (SK)

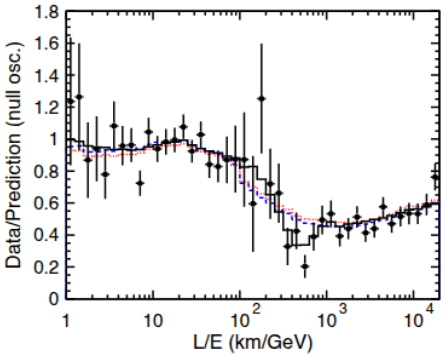
$$\sin^2(2\theta_{\rm atm}) > 0.92, \qquad \Delta m^2_{\rm atm} = (1.5\text{–}3.4) \times 10^{-3}\ {\rm eV}^2 \quad (90\%\ {\rm CL})$$

Mixing is *near-maximal*: $\theta_{\rm atm} \approx 45^\circ$.

L/E Dip and Ruling Out Neutrino Decay

- Neutrino oscillations predict a *dip* followed by a *recovery* in survival probability vs. L/E.
- Neutrino decay would give a monotonic suppression with L/E – no recovery.
- Super-Kamiokande observed the dip at $L/E \approx 500\ \mathrm{km/GeV}$, **confirming oscillations**.
- Short-baseline accelerator experiment **K2K** (KEK to Kamioka, ∼250 km, ∼1 GeV ν_μ) independently confirmed the atmospheric oscillation parameters.

Figure



Muon neutrino survival probability vs. L/E (SK).

Ruling Out Oscillation to Sterile Neutrinos

- Since ν_τ are hard to observe directly, could the ν_μ be oscillating to a **sterile neutrino** ν_s instead?
- Key: matter effects can discriminate the two scenarios.
- For $\nu_\mu \leftrightarrow \nu_\tau$: both experience the same NC potential $V_{NC} \Rightarrow$ **no net matter effect**.
- For $\nu_\mu \leftrightarrow \nu_s$: ν_s has zero weak coupling, so ν_μ sees a residual matter potential $\Rightarrow$ **matter effects present**.
- At high energies, $\Delta m^2/(2E) < A$, matter effects become large for the sterile hypothesis.
- SK observed *no such matter-induced deviation* even at high energies:

Conclusion

Oscillation to sterile neutrinos is *ruled out* as the explanation of the atmospheric neutrino anomaly. The data strongly support $\nu_\mu \to \nu_\tau$.

Effective Potentials in Matter

When neutrinos propagate through matter, they interact with ambient electrons, protons, and neutrons via the weak force.

Neutral current (NC) - all flavours equally:

$$V_{NC} = -\frac{G_F}{\sqrt{2}} N_n$$

where N_n = nucleon number density. *Common to all flavours* $\Rightarrow$ *drops out of oscillations.*

Charged current (CC) - electron neutrinos only:

$$V_{CC} = +\sqrt{2}\, G_F N_e$$

where N_e = electron number density.

Only ν_e has an additional forward-scattering amplitude off electrons via W exchange. This is the **Mikheyev-Smirnov-Wolfenstein (MSW)** effect.

Oscillations in Constant Density Matter

In a two-flavour (ν_e, ν_x) system, the effective Hamiltonian in the flavour basis becomes:

$$H_f = \frac{1}{4E}\begin{pmatrix} -\Delta m^2 \cos 2\theta + 2A & \Delta m^2 \sin 2\theta \\ \Delta m^2 \sin 2\theta & +\Delta m^2 \cos 2\theta \end{pmatrix}$$

where $A \equiv 2E\, V_{CC} = 2\sqrt{2}\, G_F N_e E$.

Diagonalising gives **effective parameters in matter**:

$$\Delta m_m^2 = \sqrt{(\Delta m^2 \cos 2\theta - A)^2 + (\Delta m^2 \sin 2\theta)^2}$$

$$\tan 2\theta_m = \frac{\Delta m^2 \sin 2\theta}{\Delta m^2 \cos 2\theta - A}$$

Oscillation formula in matter: same form as vacuum with $\Delta m^2 \to \Delta m_m^2$, $\theta \to \theta_m$.

MSW Resonance Condition

Resonance

When $A = \Delta m^2 \cos 2\theta$:

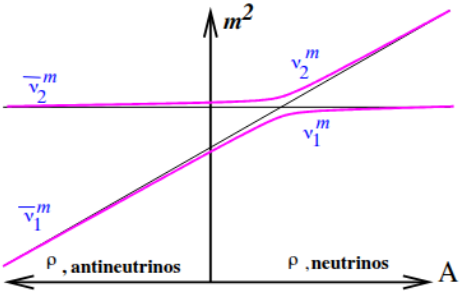
$$\tan 2\theta_m \to \infty \quad \Rightarrow \quad \theta_m = 45^\circ$$

Mixing is *maximal in matter*, *even if* θ in vacuum is small!

- Resonance enhances oscillations dramatically inside dense media (e.g. the Sun).
- For $A \ll \Delta m^2 \cos 2\theta$: $\theta_m \approx \theta$ (vacuum limit).
- For $A \gg \Delta m^2$: $\theta_m \approx 90^\circ$, mixing suppressed $\Rightarrow$ "matter suppresses mixing at very high density."
- The level-crossing diagram shows how the effective mass eigenstates evolve with density: at high densities, ν_e coincides with the heavier matter eigenstate ν_{2m}.

Level Crossing Diagram

- The effective mass-squared eigenvalues in matter as functions of A (a proxy for density).
- At $A < 0$ (antineutrinos): no resonance for $\Delta m^2 > 0$ and $\theta < 45°$.
- At $A > 0$ (neutrinos): resonance crossing occurs.
- For very high densities ($A \gg \Delta m^2$): $\nu_e \approx \nu_{2m}$ (heavier eigenstate).
- For vacuum ($A = 0$): $\nu_e \approx \cos\theta\, \nu_1 + \sin\theta\, \nu_2$.



Level crossing diagram. MSW resonance occurs where the two eigenvalue curves approach each other.

Propagation in Varying Density – Adiabatic Approximation

In the Sun or Earth, density varies with position. The local eigenstates ν_{1m}, ν_{2m} (defined by $\theta_m(x)$) also vary.

In the local eigenstate basis:

$$i\frac{d}{dt}\begin{pmatrix}\nu_{1m}\\ \nu_{2m}\end{pmatrix} = \begin{pmatrix} m_{1m}^2/2E & -i\dot{\theta}_m \\ i\dot{\theta}_m & m_{2m}^2/2E\end{pmatrix}\begin{pmatrix}\nu_{1m}\\ \nu_{2m}\end{pmatrix}$$

The off-diagonal $\dot{\theta}_m$ terms cause transitions between matter eigenstates.

Adiabatic condition (off-diagonal $\ll$ diagonal gap):

$$2\dot{\theta}_m \ll \frac{|\Delta m_m^2|}{4E}$$

At the resonance (where Δm_m^2 is smallest), the adiabaticity parameter is:

$$\gamma_{\rm res} = \frac{\Delta m^2 \sin^2 2\theta}{2E\cos 2\theta}\left|\frac{1}{A}\frac{dA}{dx}\right|_{\rm res}^{-1} \gg 1$$

If $\gamma_{\rm res} \gg 1$: neutrinos remain in the same instantaneous eigenstate (adiabatic).

Landau-Zener Jump Probability

If the transition is *non-adiabatic* at the resonance, the probability of jumping between eigenstates is given by the **Landau-Zener formula**:

$$\boxed{P_{\text{jump}} = e^{-\pi\gamma_{\text{res}}/2}}$$

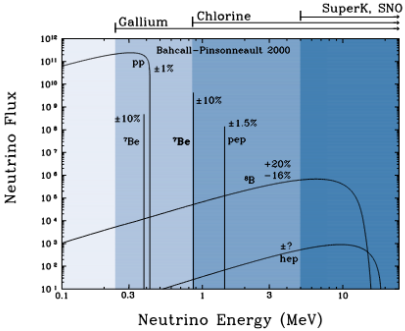
- Valid for a *linearly* varying matter potential A through the resonance.
- Shown to be a good approximation for a wide range of density profiles.
- Depends on Δm^2, θ, neutrino energy E, and the *gradient of the density profile* at the resonance.
- $\gamma_{\text{res}} \gg 1 \Rightarrow P_{\text{jump}} \to 0$: adiabatic, neutrino stays in ν_{2m}.
- $\gamma_{\text{res}} \ll 1 \Rightarrow P_{\text{jump}} \to 1$: completely non-adiabatic.

The Standard Solar Model and Solar Neutrino Production

- The Sun generates energy via nuclear fusion:

$$4p \to {}^4\mathrm{He} + 2e^+ + 2\nu_e$$

- Various reactions in the pp-chain and CNO cycle produce ν_e with characteristic energy spectra.
- Energies: $E_\nu \sim 0.1$–15 MeV.
- The **Standard Solar Model (SSM)** predicts the absolute flux and shape of each component.
- The electroweak theory fixes the *spectral shapes*; SSM provides the *normalisations*.



Solar neutrino flux vs. energy from SSM (pp, ^{7}Be, ^{8}B, ...).

Radiochemical and Water Cherenkov Detection

Radiochemical experiments (integrated flux over threshold):

$$\text{Gallium:} \quad {}^{71}\mathrm{Ga} + \nu_e \to {}^{71}\mathrm{Ge} + e^- \quad (E_{\mathrm{thr}} = 233\ \mathrm{keV})$$
$$\text{Chlorine:} \quad {}^{37}\mathrm{Cl} + \nu_e \to {}^{37}\mathrm{Ar} + e^- \quad (E_{\mathrm{thr}} = 814\ \mathrm{keV})$$

Radioactive products extracted periodically (lifetimes of weeks) and counted.

Water Cherenkov (Kamiokande, Super-Kamiokande):

$$\nu_e + e^- \to \nu_e + e^- \quad (E_{\mathrm{thr}} \approx 5\ \mathrm{MeV})$$

Scattered electron direction points back to the Sun.

Observation

Combined experiments: detected ν_e flux $\approx \mathbf{1/2}$ to $\mathbf{1/3}$ of SSM prediction. Changing flux ratios between reactions does not resolve the discrepancy $\Rightarrow$ the **solar neutrino problem**.

MSW Solutions to the Solar Neutrino Problem

Neutrinos are produced deep in the solar core where N_e is large $\Rightarrow A \gg \Delta m^2 \Rightarrow \nu_e \approx \nu_{2m}$ (heavy matter eigenstate).

As they travel outward, density decreases. Using the level-crossing picture:

- If the resonance is *adiabatic* ($\gamma_{\rm res} \gg 1$, $P_{\rm jump} \approx 0$): ν_e exits as ν_2 (vacuum eigenstate).
- In vacuum (Earth): $P(\nu_e) = |\langle \nu_e | \nu_2 \rangle|^2 = \sin^2\theta$.

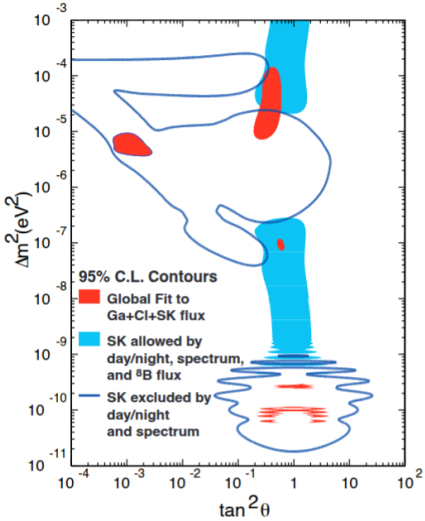
General formula:

$$P(\nu_e \to \nu_e) = P_{\rm jump} \cos^2\theta + (1 - P_{\rm jump}) \sin^2\theta$$

Four historically allowed solutions (from Δm^2, $\tan^2\theta$ fits):

Solution	Δm^2 (eV^2)	$\tan^2\theta$	Status
LMA	$\sim 10^{-4}$	~ 0.4	**Confirmed**
SMA	$\sim 10^{-5}$	$\ll 1$	Ruled out
LOW	$\sim 10^{-7}$	~ 1	Ruled out
VAC	$\sim 10^{-10}$	~ 1	Ruled out

Solar Neutrino Oscillation Solutions (2001 fit)



Allowed regions in (Δm^2, $\tan^2\theta$) plane.

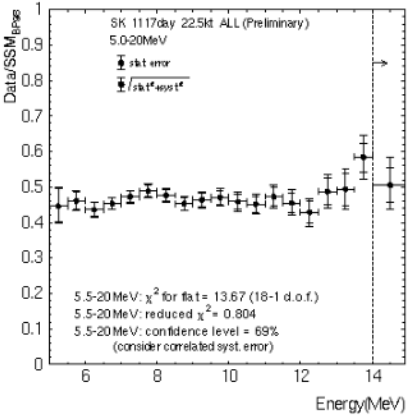
- LMA: $\Delta m^2 \sim 10^{-6}$, large θ
- SMA: $\Delta m^2 \sim 10^{-6}$, small θ
- LOW: $\Delta m^2 \sim 10^{-7}$, large θ
- VAC: $\Delta m^2 \sim 10^{-10}$, large θ

Note: this figure is from 2001; updated fits shrink all regions dramatically and eliminate SMA/LOW/VAC.

Ruling Out SMA, LOW, and VAC Solutions

- **SMA solution:** Predicts a significant *day-night asymmetry* (Earth matter effects during night-time for the resonance condition). Not observed by SK.
- **VAC solution:** Oscillations occur in vacuum between Sun and Earth. Predicts a *seasonal variation* in flux due to Earth's orbital eccentricity ($\Delta L \approx 3\%$). Not observed.
- **Energy dependence:**
  - VAC: $P_{\text{jump}} \propto \sin^2(\Delta m^2 L/E)$ - strong E-dependence.
  - SMA: P_{jump} in Landau-Zener depends on E - energy dependence.
  - LMA: $P_{\text{jump}} \approx 0$ for $E > 5$ MeV $\Rightarrow$ *flat survival probability*.
- Data show **no energy dependence for $E > 5$ MeV** $\Rightarrow$ **LMA solution is the unique survivor**.

: Solar Neutrino Survival Probability vs. Energy



The data show a flat $P(\nu_e)$ for $E > 5$ MeV, consistent with the LMA prediction ($P_{\rm jump} = 0$, i.e., $P = \sin^2\theta \approx 0.3$). Both SMA and VAC predict strong energy dependence - excluded.

Confirming Oscillations: SNO and Heavy Water

The Sudbury Neutrino Observatory (SNO) used heavy water (D_2O) and detected solar neutrinos via *three* reactions:

1. **Elastic Scattering (ES):** $\nu + e^- \to \nu + e^-$
   Sensitive to $\Phi_e + \Phi_{\mu+\tau}/6$ (cross section for ν_e is $\approx 6\times$ larger).
2. **Charged Current (CC):** $\nu_e + d \to p + p + e^-$
   Sensitive *only* to Φ_e.
3. **Neutral Current (NC):** $\nu + d \to n + p + \nu$
   Flavour-blind $\Rightarrow$ measures *total* flux $\Phi_e + \Phi_{\mu+\tau}$.

SNO results

NC measurement agreed with SSM prediction for the *total* flux. CC measurement confirmed the deficit of ν_e. $\Rightarrow$ all missing ν_e were converted to ν_μ or ν_τ, not to sterile neutrinos. $\Rightarrow$ **SSM was vindicated**.

Confirming Solar Parameters on Earth: KamLAND

- $\bar{\nu}_e$ from many Japanese nuclear reactors; baseline $\langle L \rangle \approx 180$ km; $E_\nu \sim$ few MeV.
- With LMA parameters: oscillation length $\lambda \sim 100$ km in vacuum.
- **KamLAND** (liquid-scintillator detector, ~ 1 kton) measured:[6]

$$\Delta m^2_{21} = (7.59 \pm 0.21) \times 10^{-5}\ \mathrm{eV}^2$$

$$\tan^2\theta_{12} = 0.47^{+0.06}_{-0.05}$$

- Unambiguous confirmation of the **LMA solution**.
- Physics: ν_e produced as ν_{2m} in solar core, emerge as ν_2, arrive at Earth as ν_2, detected as ν_e with probability $\sin^2\theta$.

LMA picture

Core: $A \gg \Delta m^2 \Rightarrow \nu_e \approx \nu_{2m}$

Adiabatic transition through resonance

Surface: $A \approx 0 \Rightarrow \nu_{2m} \to \nu_2$

Vacuum propagation to Earth

$P(\nu_e) = \sin^2\theta \approx 0.3$

The PMNS Mixing Matrix

Three flavour eigenstates $(\nu_e, \nu_\mu, \nu_\tau)$ are related to three mass eigenstates (ν_1, ν_2, ν_3) by:

$$\nu_\alpha = \sum_{i=1}^{3} U_{\alpha i}\, \nu_i$$

The **Pontecorvo-Maki-Nakagawa-Sakata (PMNS)** matrix (ignoring CP violation):

$$U = R_{23}(\theta_{23})\, R_{13}(\theta_{13})\, R_{12}(\theta_{12})$$

$$U = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13} & c_{12}c_{23} - s_{12}s_{23}s_{13} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13} & -c_{12}s_{23} - s_{12}c_{23}s_{13} & c_{23}c_{13} \end{pmatrix}$$

Three real parameters: θ_{12} (solar angle), θ_{23} (atmospheric angle), θ_{13} (reactor angle). With CP violation: one additional Dirac phase δ_{CP}.

General Three-Flavour Oscillation Formula

The transition probability $\nu_\alpha \to \nu_\beta$ in vacuum:

$$P_{\alpha\beta} = \delta_{\alpha\beta} - 4\sum_{i>j} \mathrm{Re}(U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i}) \sin^2\left(\frac{\Delta m^2_{ij} L}{4E}\right)$$
$$+ 2\sum_{i>j} \mathrm{Im}(U^*_{\beta j} U_{\alpha j} U_{\beta i} U^*_{\alpha i}) \sin\left(\frac{\Delta m^2_{ij} L}{2E}\right)$$

- The imaginary part (last term) is proportional to the *Jarlskog invariant* J, which is zero if CP is conserved.
- With CP conservation (real U): last term vanishes.
- Survival probability: $P_{\alpha\alpha} = 1 - 4\sum_{i>j} |U_{\alpha i}|^2 |U_{\alpha j}|^2 \sin^2\left(\frac{\Delta m^2_{ij} L}{4E}\right)$

Mass Spectrum and Hierarchies

Known from solar and atmospheric data:

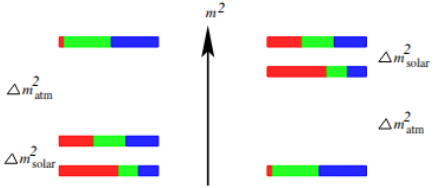
$$|\Delta m^2_{21}| \approx 8 \times 10^{-5}\ \mathrm{eV}^2 \ll |\Delta m^2_{31}| \approx 2 \times 10^{-3}\ \mathrm{eV}^2$$

Two close masses (ν_1, ν_2 separated by Δm^2_{21}) and one further away (ν_3, separated by Δm^2_{31}).

The **sign of** Δm^2_{31} is *not* determined by atmospheric/solar experiments $\Rightarrow$ two possible mass orderings:

- **Normal hierarchy (NH):** $m_1 < m_2 < m_3$
- **Inverted hierarchy (IH):** $m_3 < m_1 < m_2$

Determining the hierarchy is a key goal of future experiments.



Normal (left) and inverted (right) hierarchies. Colors show ν_e (red), ν_μ (green), ν_τ (blue) content.

Atmospheric Oscillations in the Three-Neutrino Framework

Survival probability of ν_μ:

$$P_{\mu\mu} = 1 - 4|U_{\mu 1}|^2|U_{\mu 2}|^2 \sin^2\left(\frac{\Delta m_{21}^2 L}{4E}\right) - 4|U_{\mu 2}|^2|U_{\mu 3}|^2 \sin^2\left(\frac{\Delta m_{32}^2 L}{4E}\right) - 4|U_{\mu 1}|^2|U_{\mu 3}|^2 \sin^2\left(\frac{\Delta m_{31}^2 L}{4E}\right)$$

Since $\Delta m_{21}^2 \ll \Delta m_{32}^2$, the first term is negligible for atmospheric baselines. Using $\Delta m_{31}^2 \approx \Delta m_{32}^2$ and unitarity:

$$P_{\mu\mu} \approx 1 - 4|U_{\mu 3}|^2(1 - |U_{\mu 3}|^2) \sin^2\left(\frac{\Delta m_{32}^2 L}{4E}\right)$$

Since $\cos\theta_{13} \approx 1$: $4|U_{\mu 3}|^2(1 - |U_{\mu 3}|^2) \approx \sin^2(2\theta_{23})$. Therefore:

$$\boxed{\Delta m_{\rm atm}^2 \approx |\Delta m_{31}^2| \approx 2 \times 10^{-3}\ {\rm eV}^2, \quad \theta_{\rm atm} \approx \theta_{23} \approx 45^\circ}$$

Solar Oscillations in the Three-Neutrino Framework

Survival probability of ν_e:

$$P_{ee} = 1 - 4|U_{e1}|^2|U_{e2}|^2 \sin^2\left(\frac{\Delta m_{21}^2 L}{4E}\right) - \dots$$
$$\approx 1 - 4|U_{e3}|^2(1 - |U_{e3}|^2)\sin^2\left(\frac{\Delta m_{31}^2 L}{4E}\right)$$

Key insight

Because $\theta_{13} \approx 0$ and $\Delta m_{21}^2 \ll \Delta m_{31}^2$, the solar and atmospheric systems are *decoupled* to a good approximation. Each provides clean access to one pair of mixing parameters.

Measurement of θ_{13} - CHOOZ and Reactor Experiments

- θ_{13} governs ν_e disappearance at baselines $L \sim 1$ km (driven by Δm^2_{31}):

$$P_{ee} \approx 1 - \sin^2(2\theta_{13}) \sin^2\left(\frac{\Delta m^2_{31} L}{4E}\right)$$

- **CHOOZ** (reactor $\bar{\nu}_e$, $L \approx 1$ km): no disappearance observed $\Rightarrow$ [7]

$$\sin^2(2\theta_{13}) < 0.05 \quad (3\sigma)$$

- **Daya Bay, RENO, Double CHOOZ** (2012): measured $\theta_{13} \neq 0$:

$$\sin^2(2\theta_{13}) \approx 0.085 \quad \text{(Daya Bay)}$$

- Measurement of θ_{13} is also essential for:
  - Determining the *mass hierarchy* (via matter effects in long-baseline experiments).
  - Measuring *CP violation* (δ_{CP}) in the neutrino sector.

Summary of Oscillation Parameters

Parameter	Best-fit value	Source
Δm^2_{21}	7.5×10^{-5} eV^2	Solar + KamLAND
$\lvert\Delta m^2_{31}\rvert$	2.5×10^{-3} eV^2	Atmospheric + accelerator
$\sin^2\theta_{12}$	0.30	Solar + KamLAND
$\sin^2\theta_{23}$	0.51 (near maximal)	Atmospheric + accelerator
$\sin^2\theta_{13}$	0.022	Reactor experiments
δ_{CP}	$\sim -90^\circ$ (hint)	T2K, NOvA (ongoing)

- Mass ordering: *not yet definitively determined*.
- CP violation: current data hint at a near-maximal phase, but more data needed.
- Absolute mass scale: constrained by cosmology ($\sum m_i < 0.12$ eV, Planck 2018) and KATRIN ($m_{\nu_e} < 0.8$ eV).

Overview of Neutrino Detection Reactions

Neutrinos are detected through the products of their interactions with matter. Three main channels:

1. **Charged Current (CC):** $\nu_\alpha + N \to \alpha + X$
   Neutrino converts to its partner charged lepton; lepton detected by ionisation/Cherenkov radiation.
2. **Neutral Current (NC):** $\nu_\alpha + N \to \nu_\alpha + X$
   Neutrino scatters off nucleon, hadronic final state detected.
3. **Elastic Scattering (ES):** $\nu_\alpha + e^- \to \nu_\alpha + e^-$
   Recoiling electron detected; direction points back to source.

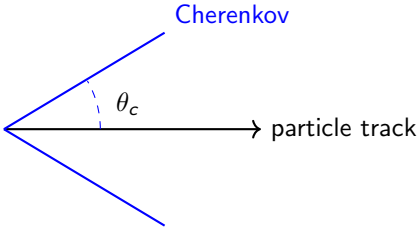
High-energy ($\gtrsim$ TeV) astrophysical neutrinos are primarily detected via Cherenkov light in large transparent natural media (water/ice).

Basics of Cherenkov Radiation

- When a charged particle moves through a medium with speed $v > c/n$ (where $n =$ refractive index), it emits **Cherenkov radiation**.
- Emission angle:

$$\cos\theta_c = \frac{1}{\beta n}, \quad \beta = v/c$$

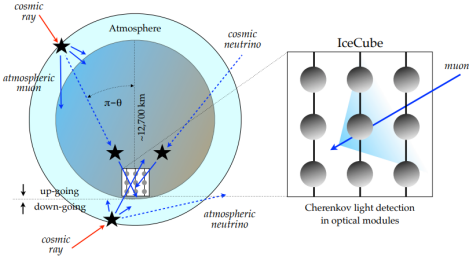
- Cherenkov light is emitted in a cone.
- Intensity $\propto \sin^2\theta_c \Rightarrow$ increases with speed.
- Detected by photomultiplier tubes (PMTs).
- For ice/water: $n \approx 1.33$, threshold $\beta > 0.75$.



Cherenkov cone: opening half-angle θ_c.

IceCube Neutrino Observatory[8]

- Located at the South Pole; 1 km^3 of ultra-transparent Antarctic ice.
- **80 strings**, each with 60 digital optical modules (DOMs), spaced 17 m apart.
- Deepest DOM at $\sim$ 2.45 km depth - shielded from cosmic-ray muon background by $\sim$ 1.5 km of ice.
- Strings at vertices of equilateral triangles, 125 m spacing.
- Each DOM: 10-inch PMT + onboard digitiser; signals time-stamped to $<$ 3 ns precision.
- Transmitted to surface for global triggering and event reconstruction.



IceCube concept at the South Pole. Upward-going muon tracks from ν_μ that have traversed the Earth are the primary cosmic neutrino signal.

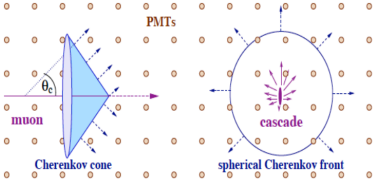
Interaction Topologies: Tracks and Cascades

Tracks (from ν_μ CC):

- Long-lived muon leaves a Cherenkov cone extending kilometers.
- Excellent angular resolution ($< 0.4^\circ$) for point-source searches.
- Disadvantage: muon energy measured only inside detector (sampling); indirect neutrino energy reconstruction (factor of ~ 2 uncertainty).
- Large atmospheric muon background $\Rightarrow$ look for upward-going tracks.

Cascades (from ν_e, ν_τ CC; all-flavour NC):

- Compact spherical light emission ($\lesssim 10$ m for PeV).
- Excellent energy resolution (total energy deposited).
- Poorer angular resolution ($\sim 8^\circ$).
- Lower atmospheric ν_e background above ~ 100 TeV.



Track (top) vs. cascade (bottom) topologies in IceCube.

Energy Measurement in IceCube

For cascades (ν_e, ν_τ):

- Total Cherenkov light $\propto$ energy deposited.
- Calorimetric measurement $\Rightarrow$ good energy resolution.
- Often fully (or partially) contained in detector.

For tracks (ν_μ):

- Muon loses energy gradually: ionisation, bremsstrahlung, pair production, photo-nuclear interactions.
- Energy-loss cone is km-long with *decreasing radius* (Cherenkov cone narrows as muon loses energy).
- Only the energy deposited *inside* the detector is observed; an unfolding procedure recovers the initial neutrino energy.
- Typical precision: factor of ~ 2 in energy, but systematic uncertainties cancel in spectrum measurements.
- At hundreds of GeV and below: muon becomes minimum-ionising.

Key threshold

Above $\sim$ 300 TeV, atmospheric neutrinos are rare even in 1 km^3 $\Rightarrow$ every event is likely of *cosmic* origin.

Angular Resolution and Background Rejection

Tracks:

- Angular resolution $< 0.4^\circ \Rightarrow$ ideal for point-source searches.
- Background: (a) atmospheric cosmic-ray muons (downgoing) $\Rightarrow$ select upward-going only; (b) atmospheric ν_μ (upgoing) $\Rightarrow$ irreducible below ~ 100 TeV.
- At very high energies: ν and μ are nearly co-aligned $\Rightarrow$ sensitivity to point sources maximised.

Cascades:

- Angular resolution $\sim 8^\circ$ with current techniques; ongoing improvement via better ice characterisation and calibration.
- Detected from both hemispheres (no Earth shadowing).
- Atmospheric ν_e background lower by > 1 order of magnitude above 100 TeV (pions freeze out; K decays dominate, but rare).
- ν_τ essentially *not absorbed* by Earth: $\nu_\tau \to \tau \to \nu_\tau$ cascade-down, re-emerges at lower energy.

Cosmic Neutrino Discovery and Landau-Pomeranchuk-Migdal Effect

- IceCube announced the first detection of a diffuse flux of astrophysical neutrinos in 2013, observed as an excess of cascade events above 30 TeV.
- The cosmic neutrino spectrum is approximately $\propto E^{-2}$, while atmospheric is $\propto E^{-3.7}$ $\Rightarrow$ spectral separation.
- One ν_τ event at $\sim$ 100 TeV provides an independent confirmation of cosmic neutrinos.

Landau-Pomeranchuk-Migdal (LPM) effect above $\sim$ 100 PeV

Electromagnetic showers begin to *elongate* because soft photons from multiple centres interfere destructively with single-atom scattering, reducing energy loss. This changes the shower topology at the highest energies and must be accounted for in reconstruction.

Open Questions and Future Experiments

Key unknowns:

- **Mass ordering**: normal or inverted?
- **CP violation**: is δ_{CP} large? Is there leptogenesis?
- **Absolute mass scale**: direct measurement below 0.1 eV?
- **Dirac or Majorana**: $0\nu\beta\beta$ searches.
- **Sterile neutrinos**: hints from LSND, MiniBooNE – unresolved.
- **Cosmic sources**: what accelerates PeV neutrinos?

Future experiments:

- **JUNO** (reactor, China): mass ordering, precision Δm^2_{21} and θ_{12}.
- **DUNE** (FNAL→SURF, 1300 km): mass ordering + CP violation.
- **Hyper-Kamiokande**: CP violation, proton decay.
- **KM3NeT**: Mediterranean km^3 detector.
- **IceCube-Gen2**: 10 km^3, extended energy reach.
- **Project 8**: tritium m_ν via cyclotron radiation.

Conclusion

The neutrino sector in one paragraph

Neutrinos were introduced as a bookkeeping device to save energy conservation in beta decay. They turned out to be the most elusive, yet most prevalent, fermions in the universe. Decades of ingenious experiments – from reactor detectors to cubic-kilometre of ice – have revealed that neutrinos *oscillate*, have *mass*, and mix with large angles, in striking contrast to the quark sector. Every confirmed oscillation result is a window beyond the Standard Model. The programme of neutrino physics is far from over.

Thank you!

"The neutrino is the most tiny quantity of reality ever imagined by a human being."

— *Frederick Reines*

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[8] Francis Halzen and Ali Kheirandish. "IceCube and high-energy cosmic neutrinos". In: *The Encyclopedia of Cosmology: Set 2: Frontiers in Cosmology Volume 2 Neutrino Physics and Astrophysics*. World Scientific, 2024, pp. 107–235.

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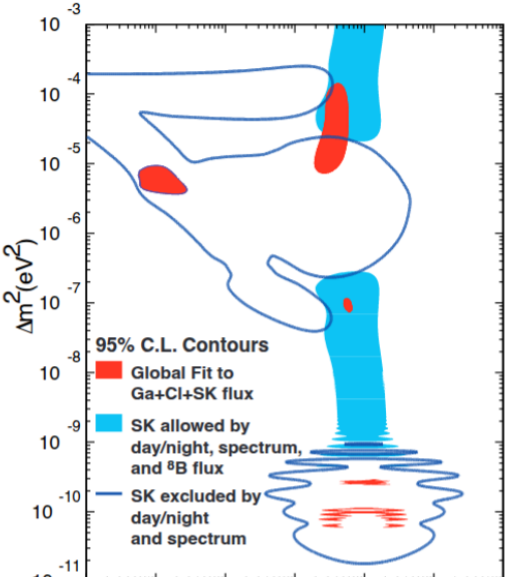
Backup: Solar Neutrino Oscillation Solutions (2001 fit)



Allowed regions in (Δm^2, $\tan^2\theta$) plane.

- LMA: $\Delta m^2 \sim 10^{-6}$, large θ
- SMA: $\Delta m^2 \sim 10^{-6}$, small θ
- LOW: $\Delta m^2 \sim 10^{-7}$, large θ
- VAC: $\Delta m^2 \sim 10^{-10}$, large θ

Note: this figure is from 2001; updated fits shrink all regions dramatically and eliminate SMA/LOW/VAC.

Backup: Three-Neutrino Mixing Parameters (Current Values)

Parameter	Normal Ordering	Inverted Ordering
$\sin^2\theta_{12}$	0.304	0.304
$\sin^2\theta_{23}$	0.573	0.575
$\sin^2\theta_{13}$	0.02219	0.02238
δ_{CP} $(^\circ)$	197^{+27}_{-24}	286^{+27}_{-32}
Δm^2_{21} $(10^{-5}$ eV$^2)$	7.42	7.42
$\lvert\Delta m^2_{31}\rvert$ $(10^{-3}$ eV$^2)$	2.517	2.498

(NuFIT 5.3, 2023 - global analysis [9], these values are accurate upto $\mathcal{O}$ (10%) at 3 σ)

Part II: Neutrino Mass Mechanisms – Roadmap

What follows

From here, we move beyond oscillations and focus on the **origin of neutrino masses**:

- Dirac vs. Majorana neutrinos
- Neutrinoless double-beta decay ($0\nu\beta\beta$)
- Weinberg dimension-5 operator
- Seesaw mechanisms (Type I, II, III)
- Sterile neutrinos
- Leptogenesis
- Cosmological constraints ($\sum m_\nu$, $N_{\rm eff}$)
- Mass hierarchy from DESI+Planck

These ideas collectively try to explain why neutrinos have mass and how they connect to the early universe.

The Standard Model Paradigm

- In the Standard Model (SM), neutrinos are fundamentally chiral; we only observe **left-handed neutrinos** (ν_L).
- They belong to the $SU(2)_L$ lepton doublet: $L = (\nu_L, \ell_L)^T$.
- They transform as $(\mathbf{1}, \mathbf{2}, -1/2)$ under the SM gauge group.
  - *Notation: (color singlet, weak doublet, hypercharge Y).*

The Particle Gap

Crucially, there are **no right-handed neutrinos (ν_R)** present in the Standard Model particle content.

The Mass Generation Problem in the SM

- Generating a fermion mass requires coupling a left-handed field to a right-handed partner.
- Since left-handed neutrinos lack right-handed partners, they cannot form this standard coupling.
- Furthermore, there is **no other renormalizable term** in the SM that can generate mass via the Higgs Vacuum Expectation Value (VEV).

Theoretical Conclusion

In the minimal Standard Model, neutrinos are identically **massless**.

Empirical Evidence for Neutrino Mass

- Oscillation experiments proved that flavor states mix, requiring physical mass eigenstates (m_1, m_2, m_3).
- We have measured two distinct, non-zero mass squared differences:

Measured Mass Splittings

$$\Delta m_{21}^2 \equiv m_2^2 - m_1^2 \approx 7.5 \times 10^{-5} \ \mathbf{eV}^2$$
$$|\Delta m_{31}^2| \equiv |m_3^2 - m_1^2| \approx 2.5 \times 10^{-3} \ \mathbf{eV}^2$$

- **The Crisis:** Because these splittings are non-zero, at least two neutrinos **must** possess a non-zero absolute mass.

Solving the Crisis: Two Fundamental Methods

To accommodate mass, we must look beyond the Standard Model. We have two primary mathematical methods to generate this mass:

1. **The Dirac Method:** Introduce entirely new particles to create a standard, **renormalizable** mass term.
2. **The Majorana Method:** Construct a mass term using only existing fields, which requires **non-renormalizable**, high-scale physics.

Method 1: The Dirac Mass

- We introduce right-handed neutrinos (ν_R) as total singlets: $(\mathbf{1}, \mathbf{1}, 0)$.
- This allows us to write a standard Yukawa interaction using the conjugate Higgs doublet ($\tilde{\phi} = i\tau_2\phi^*$):

Dirac Mass Lagrangian

$$\mathcal{L}_{\text{Yuk}} = -y_\nu \, \bar{L}\tilde{\phi}\nu_R + \text{h.c.}$$

- After EWSB ($\langle\phi\rangle = v/\sqrt{2}$), we get a Dirac mass: $m_D = y_\nu v/\sqrt{2}$.

The Dirac Naturalness Problem

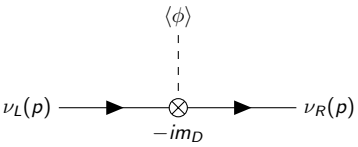


Figure: Dirac neutrino mass insertion.

- This method perfectly **conserves** total lepton number (L).
- **Naturalness Issue:** Since $m_D \lesssim 0.1$ eV and $v \approx 246$ GeV:
- The required Yukawa coupling is $y_\nu \sim 10^{-12}$.
- This extreme fine-tuning motivates us to explore the Majorana alternative.

Method 2: Majorana Neutrinos

- Ettore Majorana (1937) demonstrated that a neutral fermion can be its own antiparticle.
- We construct a right-handed chiral state from our existing ν_L using the charge conjugation matrix $C = i\gamma^2\gamma^0$:

$$\nu_L^c \equiv C\overline{\nu_L}^T$$

- Since this state transforms as a right-handed spinor, we can couple it directly to ν_L to form a Lorentz scalar.

The Majorana Mass Term

Majorana Lagrangian

$$\mathcal{L}_{\mathrm{Maj}} = -\frac{1}{2} m_M \left(\overline{\nu_L^c} \nu_L + \overline{\nu_L} \nu_L^c \right) = -\frac{1}{2} m_M \, \nu_L^T C^{-1} \nu_L + \text{h.c.}$$

- **Lepton Number Violation:** Coupling two left-handed fields explicitly violates total lepton number by two units ($\Delta L = 2$).
- **Isospin Constraint:** The bilinear $\nu_L^T C^{-1} \nu_L$ has weak isospin $I_3 = 1$, which strictly requires new, high-scale physics beyond EWSB.

Majorana Mass Insertion ($\Delta L = 2$)

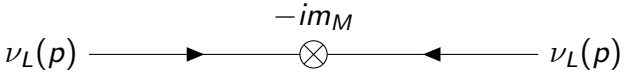


Figure: Majorana mass insertion. Clashing fermion arrows demonstrate the annihilation/creation of two left-handed neutrinos.

The Majorana Mass: A High-Scale Necessity

- A Majorana mass must arise from a high-scale Λ as a non-renormalizable operator:

$$\mathcal{L}_{\text{eff}} = \frac{c}{\Lambda}(\bar{L}\tilde{\phi})(\tilde{\phi}^T L^c) + \text{h.c.}$$

- After symmetry breaking: $m_\nu = c\, v^2/\Lambda$.
- Observed $m_\nu \sim 0.1$ eV implies $\Lambda \sim 3 \times 10^{14}$ **GeV**.

The Scale of New Physics

This scale points directly toward the **Grand Unification (GUT) scale**.

Connecting to Seesaw Mechanisms

- The effective dimension-5 operator is the low-energy footprint of a heavier, renormalizable theory.
- The three **Seesaw Mechanisms** (Types I, II, III) are the unique tree-level UV completions of this operator.
- They introduce heavy particles at the scale Λ which, when integrated out, generate the suppression $m_\nu \sim v^2/M_{\text{heavy}}$.
- **Outlook:** These types naturally solve mass smallness while predicting testable phenomena like $0\nu\beta\beta$ and leptogenesis.

Phenomenological Difference: Lepton Number

- Dirac and Majorana neutrinos are **kinematically indistinguishable** in oscillations.
- The fundamental difference is the conservation of total lepton number L:

Dirac Neutrinos

$\Delta L = 0$ **(Conserved)**
Neutrinos and antineutrinos are distinct.

Majorana Neutrinos

$\Delta L = 2$ **(Violated)**
Neutrinos are their own antiparticles.

From $\Delta L = 2$ to Neutrinoless Double-Beta Decay

- Violation of lepton number allows processes like $0\nu\beta\beta$:

$$(A, Z) \to (A, Z + 2) + 2e^-$$

- No neutrinos are emitted; $\Delta L = 2$ is carried by the Majorana propagator.

The "Smoking Gun"

$0\nu\beta\beta$ is strictly forbidden for Dirac neutrinos. Its observation would be definitive proof of the Majorana nature of neutrinos. (Proof later)

Comparing Double-Beta Decay Channels

Two-neutrino double-beta decay ($2\nu\beta\beta$)

$$\mathcal{N}(A, Z) \to \mathcal{N}(A, Z+2) + 2e^- + 2\bar{\nu}_e$$

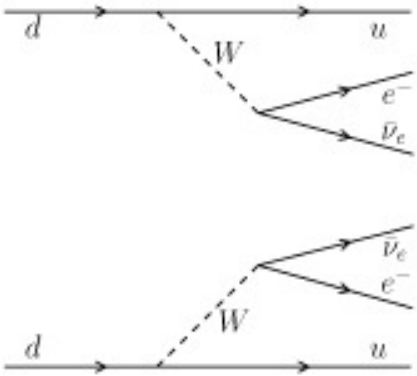
- Second-order weak process; **allowed** in the Standard Model.
- Observed in several isotopes (e.g., ^{76}Ge, ^{136}Xe).

Neutrinoless double-beta decay ($0\nu\beta\beta$)

$$\mathcal{N}(A, Z) \to \mathcal{N}(A, Z+2) + 2e^-$$

- **Forbidden** in the SM; violates Lepton Number ($\Delta L = 2$).
- No antineutrinos are emitted in the final state.

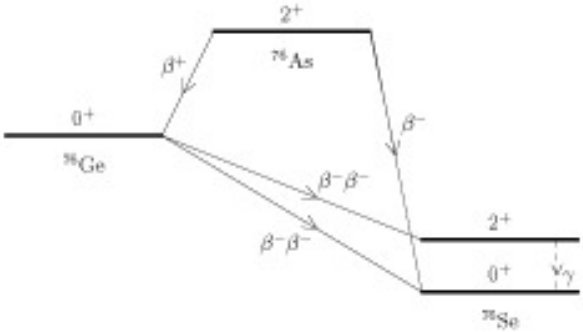
Standard Model Process: The $2\nu\beta\beta$ Mechanism



Tree-Level Quark Topology

Two neutrons decay simultaneously via two W^- boson exchanges, producing two electrons and two antineutrinos. (Adapted from Giunti & Kim)

Kinematic Requirements: The ^{76}Ge Case Study



Single β Suppression

In isotopes like ^{76}Ge, single β^- decay to ^{76}As is **kinematically forbidden** (ground state of ^{76}As is higher in energy).

- This forces the nucleus to "wait" for the much rarer double-beta transition to ^{76}Se.

Majorana Nature and Helicity Mismatch

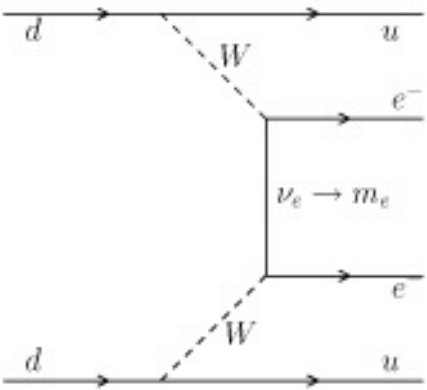
For $0\nu\beta\beta$ to occur via the standard mechanism, two fundamental mismatches must be resolved at the internal neutrino line:

1. **Particle–Antiparticle Mismatch:** The first vertex emits $\bar{\nu}_e$, but the second vertex requires absorbing ν_e.
2. **Helicity Mismatch:** The weak interaction emits a *right-handed* antineutrino, but requires a *left-handed* neutrino for absorption.

The Majorana Solution

If ν is a Majorana particle, $\bar{\nu} = \nu$ (solving mismatch 1). A non-zero mass allows a "wrong-helicity" component with amplitude $\sim m_\nu / E_\nu$ (solving mismatch 2).

Lepton Number Violation via Majorana Exchange



LNV Signature

The internal Majorana neutrino propagator connects the vertices and carries $\Delta L = 2$. (Adapted from Giunti & Kim)

The Effective Majorana Mass $m_{2\beta}$

In the three-flavor mixing framework, the decay amplitude is directly proportional to the **effective Majorana mass**:

The Master Formula

$$m_{2\beta} = \left| \sum_{k=1}^{3} U_{ek}^2 \, m_k \right|$$

- Expanding with Majorana phases α_2, α_3:

$$m_{2\beta} = \left| |U_{e1}|^2 m_1 + |U_{e2}|^2 m_2 e^{i\alpha_2} + |U_{e3}|^2 m_3 e^{i\alpha_3} \right|$$

- **Interference:** Destructive interference between terms can make $m_{2\beta}$ smaller than the lightest mass eigenstate.

Derivation of the Internal Neutrino Propagator

The transition amplitude in $0\nu\beta\beta$ is governed by the contraction of two flavor-state neutrino fields:

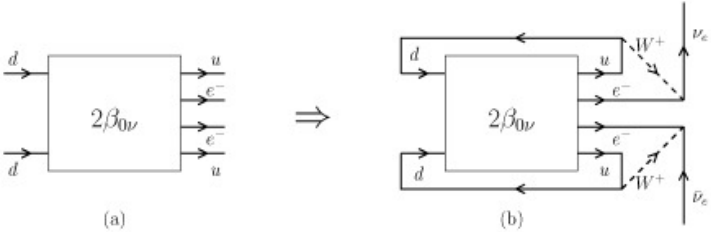
$$\begin{aligned} G(x_1 - x_2) &= \langle 0| T[\nu_{eL}(x_1)\nu_{eL}^T(x_2)]|0\rangle \\ &= \frac{1-\gamma^5}{2}\sum_{k=1}^{3} U_{ek}^2 \langle 0| T[\nu_k(x_1)\nu_k^T(x_2)]|0\rangle \left(\frac{1-\gamma^5}{2}\right)^T \\ &= -i\sum_{k=1}^{3} U_{ek}^2 \int \frac{d^4p}{(2\pi)^4}\frac{m_k}{p^2 - m_k^2} e^{-ip\cdot(x_1-x_2)}\frac{1-\gamma^5}{2}\mathcal{C} \end{aligned}$$

- **Nuclear Scale:** The average neutrino momentum is $\langle p\rangle \sim 35$ MeV, which is much larger than the mass eigenvalues $m_k < 0.1$ eV.
- **Approximation:** We can neglect m_k^2 in the denominator and factor the effective mass out of the integral.

Propagator Approximation

$$G(x_1 - x_2) \approx -i m_{2\beta} \int \frac{d^4p}{(2\pi)^4}\frac{e^{-ip\cdot(x_1-x_2)}}{p^2}\frac{1-\gamma^5}{2}\mathcal{C}$$

The Schechter–Valle "Black Box" Theorem



From Decay to Mass

Left: Generic black box for $2d \to 2u + 2e$. **Right:** By closing the loops with W bosons, any $0\nu\beta\beta$ mechanism generates a Majorana mass radiatively.

The Rigorous Guarantee of Schechter–Valle

- If $0\nu\beta\beta$ occurs via *any* new physics mechanism (e.g., SUSY, heavy scalars), it necessarily implies Majorana masses.
- The joining of external electron and quark lines with W bosons creates a 4-loop diagram for $\bar{\nu}_e \to \nu_e$.
- No symmetry can forbid this generated mass without also forbidding the initial $0\nu\beta\beta$ decay

Theoretical Conclusion

The observation of $0\nu\beta\beta$ rigorously **guarantees** that neutrinos are massive Majorana particles, regardless of whether the light-neutrino exchange is the dominant mechanism.

Extracting Mass from Half-Life Measurements

The experimental $0\nu\beta\beta$ half-life for a nucleus $\mathcal{N}$ is given by:

The Half-Life Formula

$$[T_{1/2}^{0\nu}(\mathcal{N})]^{-1} = G_{0\nu}^{\mathcal{N}} \, |\mathcal{M}_{0\nu}^{\mathcal{N}}|^2 \, m_{2\beta}^2$$

- $G_{0\nu}^{\mathcal{N}}$**:** Precisely calculable phase-space factor.
- $\mathcal{M}_{0\nu}^{\mathcal{N}}$**: Nuclear Matrix Element (NME)** – encodes complex nuclear structure.

Theoretical Uncertainty

Different nuclear models (QRPA, Shell Model, IBM) give NMEs that vary by factors of 2–3. This is currently the largest bottleneck in interpreting data.

The Experimental Frontier (2025–2026)

Current Leading Constraints

- **LEGEND-200 (^{76}Ge):** $T^{0\nu}_{1/2} > 1.9 \times 10^{26}$ yr $\Rightarrow m_{2\beta} < (75 - 200)$ meV.
- **KamLAND-Zen (^{136}Xe):** $T^{0\nu}_{1/2} > 2.3 \times 10^{26}$ yr $\Rightarrow m_{2\beta} < (36 - 156)$ meV.
- **CUORE (^{130}Te):** $T^{0\nu}_{1/2} > 3.2 \times 10^{25}$ yr.

The Ton-Scale Era

Future experiments (LEGEND-1000, nEXO) aim to reach $m_{2\beta} \sim 10$ **meV**, allowing a definitive probe of the entire Inverted Hierarchy region.

Final Comparison: Dirac vs. Majorana

Property	Dirac	Majorana
Particle content	ν_L, ν_R	only ν_L (or ν_L^c)
Degrees of freedom	4	2
Total lepton number L	Conserved	Violated ($\Delta L = 2$)
Mass term renormalizable?	Yes	No (requires $\Lambda \gg v$)
Experimental test	$0\nu\beta\beta$ absent	$0\nu\beta\beta$ present
Naturalness	$y_\nu \sim 10^{-12}$	$m_\nu \sim v^2/\Lambda$

The Path Ahead

The search for $0\nu\beta\beta$ remains the only practical way to distinguish these scenarios. The coming decade will be decisive.

Why Neutrino Masses Require Physics Beyond the Standard Model

- In the renormalizable Standard Model (SM), all fermion masses arise from **Yukawa couplings** of the form $\bar{\psi}_L \phi \psi_R$ after electroweak symmetry breaking (EWSB).
- For neutrinos, there is **no right-handed partner** ν_R in the SM particle content. Hence the usual Dirac mass term cannot be written.
- Moreover, the SM Higgs doublet has weak isospin $I_3 = -1/2$, while a **Majorana mass term** $\nu_L^T C^{-1} \nu_L$ would require $I_3 = +1$ – impossible with renormalizable interactions.

Conclusion

Neutrino masses are strictly zero in the SM. Their experimental observation forces us to extend the SM, and the extension must operate at a **high energy scale**.

Effective Field Theory: The Natural Language for New Physics

- If new physics exists at a very high scale Λ (much larger than the electroweak scale), its effects at low energies can be captured by an **effective field theory** (EFT).
- The EFT Lagrangian is an expansion in powers of $1/\Lambda$:

$$\mathcal{L}_{\text{EFT}} = \mathcal{L}_{\text{SM}} + \frac{1}{\Lambda}\sum_k c_k^{(5)}\mathcal{O}_k^{(5)} + \frac{1}{\Lambda^2}\sum_k c_k^{(6)}\mathcal{O}_k^{(6)} + \cdots ,$$

  where $\mathcal{O}_k^{(d)}$ are gauge- and Lorentz-invariant operators of dimension $d > 4$, and $c_k^{(d)}$ are dimensionless **Wilson coefficients** (expected to be $\mathcal{O}(1)$ in a generic UV completion).
- Higher-dimensional operators are suppressed by powers of $1/\Lambda$, so the lowest dimension operators give the largest low-energy effects.

Historical Precedent for Effective Field Theory

- Using non-renormalizable operators to predict high-scale physics has a famous historical precedent in weak interactions.
- **Fermi's Theory (1934):** Beta decay was originally modeled as a dimension-6, 4-fermion contact interaction:

$$\mathcal{L}_{\text{Fermi}} = \frac{G_F}{\sqrt{2}}(\bar{p}\gamma^\mu n)(\bar{e}\gamma_\mu \nu_e)$$

- At low energies, this worked perfectly. But the coupling G_F has dimensions of $[\text{Mass}]^{-2}$.
- This dimensional suppression correctly predicted that the contact interaction was actually the low-energy limit of a heavy mediator exchange: the W **boson** ($G_F \sim g^2/M_W^2$).

Modern Analogy

Today, we use the exact same logic for neutrino masses, but starting with a dimension-5 operator to predict the Seesaw mediators.

The Weinberg Operator (1979): The Unique Dimension-5 Operator

- In 1979, Steven Weinberg asked: what is the **lowest possible dimension** of an operator that can generate a Majorana neutrino mass, given the SM gauge symmetry and particle content?
- The answer: **dimension 5**. Moreover, there is exactly **one independent** such operator:

$$\mathcal{O}_5 = \frac{c_{\alpha\beta}}{\Lambda}\left(\overline{L_\alpha}\tilde{\phi}\right)\left(\tilde{\phi}^T L^c_\beta\right) + \text{h.c.}$$

- Here $L_\alpha = (\nu_\alpha, \ell_\alpha)^T$ is the left-handed lepton doublet of flavor α, $\tilde{\phi} = i\tau_2\phi^*$ is the conjugate Higgs doublet, and $L^c = C\overline{L}^T$ is the charge-conjugated field.
- The coefficient matrix $c_{\alpha\beta}$ is strictly symmetric ($c_{\alpha\beta} = c_{\beta\alpha}$). The anti-commuting nature of the fermions perfectly cancels the anti-symmetry of the charge conjugation matrix C.

Lepton Number Violation and the Isospin Argument

- Assign a global **lepton number** $L = +1$ to the lepton doublet L_α (and $L = -1$ to its antiparticle). Under a $U(1)_L$ phase rotation $L \to e^{i\alpha} L$, the Weinberg operator transforms as:

$$\mathcal{O}_5 \to e^{2i\alpha} \mathcal{O}_5.$$

- **Therefore, $\mathcal{O}_5$ violates total lepton number by two units ($\Delta L = 2$).** This is the hallmark of Majorana neutrino mass generation.
- **Weak isospin consideration:** The lepton doublet L has $I_3 = +1/2$, the conjugate Higgs $\tilde{\phi}$ has $I_3 = -1/2$. The product $(\bar{L}\tilde{\phi})$ is an isosinglet ($I = 0$). Two such isosinglets combine to give total $I = 0$, but the fermion bilinear $\bar{L}^c L$ requires $I_3 = +1$ – a mismatch.

Consequence

$\mathcal{O}_5$ cannot arise from any renormalizable interaction of the SM. It must be generated by **new heavy physics** at the scale Λ.

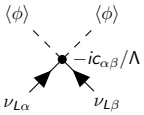
From the Weinberg Operator to a Majorana Mass

- After EWSB, the Higgs acquires a vacuum expectation value $\langle\phi\rangle = (0, v/\sqrt{2})^T$.
- Substituting this into $\mathcal{O}_5$ yields:

$$\mathcal{O}_5 \to \frac{c_{\alpha\beta}}{\Lambda}\frac{v^2}{2}\overline{\nu^c_{L\alpha}}\nu_{L\beta}.$$

- Hence the **Majorana mass matrix** for the left-handed neutrinos is:

$$\boxed{(m_\nu)_{\alpha\beta} = c_{\alpha\beta}\frac{v^2}{\Lambda}}$$



The filled circle denotes the **effective vertex**. The two external Higgs lines are frozen at their VEVs, turning the operator into a mass insertion.

Naturalness and the High-Energy Scale Λ

- In the sense of 't Hooft, the smallness of m_ν is **technically natural**: setting $m_\nu \to 0$ restores the $U(1)_L$ symmetry of the SM. Therefore, quantum corrections to m_ν are proportional to m_ν itself – no fine-tuning is needed to keep it small.
- Assuming the Wilson coefficients are of order unity ($c_{\alpha\beta} \sim \mathcal{O}(1)$), the observed neutrino mass scale $m_\nu \sim 0.1$ eV allows us to **estimate the scale of new physics**:

$$\Lambda \sim \frac{v^2}{m_\nu} \sim \frac{(174\ \text{GeV})^2}{0.1\ \text{eV}} \sim 3 \times 10^{14}\ \text{GeV}.$$

- This is remarkably close to the **Grand Unified Theory (GUT) scale** ($\sim 10^{16}$ GeV). The Weinberg operator thus points toward physics at extremely high energies, far beyond direct reach of colliders.

UV Completion

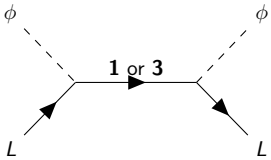
The exact UV completion – whether it involves heavy right-handed neutrinos, scalar triplets, or fermion triplets – is **abstracted away** by the EFT framework. However, the Schechter–Valle black box theorem guarantees that observation of $0\nu\beta\beta$ would prove the operator is realized in nature.

Unpacking the Operator: Why Only Three Completions?

- To find the physical origins of the 4-point Weinberg operator ($LL\phi\phi$), we must "open it up" into two 3-point vertices connected by a heavy mediator.
- **Path 1: Pair a Lepton with a Higgs ($L \otimes \phi$).**
- This creates a heavy intermediate **fermion**.
- **Group Theory:** Both L and ϕ are $SU(2)_L$ doublets ($\mathbf{2}$).

$$\mathbf{2} \otimes \mathbf{2} = \mathbf{1} \oplus \mathbf{3}$$

- The heavy fermion must be a **Singlet** (Type I) or a **Triplet** (Type III).

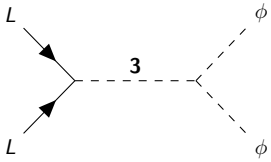


Unpacking the Operator: The Scalar Path

- **Path 2: Pair a Lepton with a Lepton ($L \otimes L$).**
- This topology creates a heavy intermediate **scalar boson** that eventually decays into two Higgs fields.
- **Group Theory:** Again, we multiply the doublets:

$$\mathbf{2} \otimes \mathbf{2} = \mathbf{1}_A \oplus \mathbf{3}_S$$

- **Fermi Statistics:** Generating a valid Majorana mass requires a symmetric matrix. Thus, we are forced to use the **symmetric Triplet** ($\mathbf{3}_S$).
- The heavy scalar must be a **Triplet** (Type II).



The Seesaw Landscape: Summary of UV Completions

- As derived, opening the Weinberg operator at tree level strictly limits us to exactly **three heavy mediators**:

The Three Tree-Level Completions

1. **Type I Seesaw (Fermion Singlet):** Exchange of heavy right-handed neutrinos $N_R \sim (\mathbf{1}, \mathbf{1}, 0)$.
2. **Type II Seesaw (Scalar Triplet):** Exchange of a heavy scalar $\Delta \sim (\mathbf{1}, \mathbf{3}, +2)$.
3. **Type III Seesaw (Fermion Triplet):** Exchange of heavy fermion triplets $\Sigma \sim (\mathbf{1}, \mathbf{3}, 0)$.

- While all three pathways naturally generate sub-eV masses, they predict radically different **collider signatures** (e.g., searching for doubly-charged scalars Δ^{++} in Type II, versus heavy lepton tracks in Type III).

Seesaw Mass Scaling: Fermionic vs. Scalar Mediators

- For **fermionic mediators** (Type I and Type III), the Weinberg operator is generated by the exchange of a heavy fermion. The diagram involves two Yukawa couplings Y and one heavy propagator. After integrating out, the light neutrino mass scales as:

$$m_\nu \sim \frac{Y^2 v^2}{M_{\text{heavy}}}.$$

- For the **scalar mediator** (Type II), the diagram involves a Yukawa coupling Y_Δ to the leptons, a trilinear coupling μ (with mass dimension) that connects the scalar triplet to two Higgs doublets, and the heavy scalar propagator. This gives:

$$m_\nu \sim \frac{\mu\, Y_\Delta v^2}{M_{\text{heavy}}^2}.$$

Common Origin

In all cases, m_ν is **suppressed by the heavy scale**, explaining its smallness naturally if $M_{\text{heavy}} \sim 10^{14}$ GeV and the couplings are of order one.

Type I Seesaw: Heavy Right-Handed Singlets

- We extend the SM by adding $n \geq 2$ **right-handed neutrinos** (N_R). As complete gauge singlets $(\mathbf{1}, \mathbf{1}, 0)$, they do not feel SM forces ("sterile").
- This singlet nature allows us to write the Type I Lagrangian:

$$\mathcal{L}_{\mathrm{I}} = \mathcal{L}_{\mathrm{SM}} + i\overline{N_R}\partial\!\!\!/ N_R - \left[(Y_\nu)_{\alpha i}\, \overline{L_\alpha}\tilde{\phi}\, N_{Ri} + \frac{1}{2}\overline{N_{Ri}^c}\, (M_R)_{ij}\, N_{Rj} + \text{h.c.} \right]$$

- **The Dirac Term (Y_ν):** A standard Yukawa coupling linking the active SM lepton doublets to the sterile N_R states via the Higgs field.
- **The Majorana Term (M_R):** Because N_R carries no weak isospin or hypercharge, a bare Majorana mass term is **explicitly allowed** by gauge invariance without requiring symmetry breaking.
- The coexistence of the electroweak Dirac mass and the super-heavy Majorana mass forms the fundamental engine of the seesaw.

Type I: The Neutrino Mass Matrix

- After electroweak symmetry breaking, the Higgs acquires its VEV $\langle\phi\rangle = (0, v/\sqrt{2})^T$, leading to the **Dirac mass matrix**:
$$m_D = Y_\nu \frac{v}{\sqrt{2}}.$$
- In the basis of left-handed chiral fields $\Psi_L = (\nu_L,\, N_R^c)^T$, the full $(3+n)\times(3+n)$ neutrino mass matrix takes the symmetric block form:
$$\mathcal{M} = \begin{pmatrix} 0 & m_D \\ m_D^T & M_R \end{pmatrix}.$$
- The zero in the upper-left block is a consequence of the fact that the SM alone does not contain a Majorana mass for ν_L (that would require a triplet Higgs). The off-diagonal blocks mix the active and sterile states, and the lower-right block is the heavy Majorana mass M_R.

Type I: The Seesaw Diagonalization

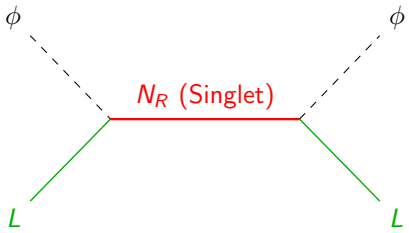
- In the **seesaw limit** where all eigenvalues of M_R are much larger than the electroweak scale ($M_R \gg m_D$), we can perform a block-diagonalization of $\mathcal{M}$ using a unitary transformation.
- The physical spectrum splits into two sectors:
  1. **Heavy sector:** n states with masses approximately equal to M_R, composed predominantly of the right-handed singlets N_R. These are too heavy to be produced at current colliders if $M_R \sim 10^{14}$ GeV.
  2. **Light sector:** three active neutrinos with masses given by the **celebrated seesaw formula**:

$$\boxed{m_\nu \simeq -m_D M_R^{-1} m_D^T = -\frac{v^2}{2} Y_\nu M_R^{-1} Y_\nu^T .}$$

Naturalness

The smallness of m_ν is **naturally explained** by the large suppression factor v^2/M_R. If $Y_\nu \sim 1$ and $M_R \sim 10^{14}$ GeV, then $m_\nu \sim 0.1$ eV – exactly what is observed.

Type I: Tree-Level Feynman Diagram



- The diagram shows the **exchange of a heavy Majorana neutrino** N_R between two SM lepton–Higgs vertices.
- At low energies ($E \ll M_R$), the propagator shrinks to a point, generating the effective Weinberg operator.
- The color coding: green for SM lepton doublets L, red for the new heavy singlet N_R.

Leptogenesis: A Consequence of the Seesaw

- All three seesaw mechanisms can naturally generate the **observed baryon asymmetry** of the universe via **leptogenesis**.
- At $T \sim M_{\text{heavy}}$, heavy mediators (N_R, Δ, or Σ) are thermally produced, then decay out of thermal equilibrium.
- These decays violate **Lepton Number** ($\Delta L = 2$) and can violate **CP** if the Yukawa matrices contain complex phases.
- Interference between tree-level and one-loop decay diagrams generates a **CP asymmetry** (ε):

$$\varepsilon \simeq \frac{1}{8\pi} \sum_{j \neq 1} \frac{\mathrm{Im}\left[(Y^\dagger Y)^2_{1j}\right]}{(Y^\dagger Y)_{11}} \left(\frac{M_1}{M_j}\right)$$

- Non-perturbative **electroweak sphalerons** then convert this lepton asymmetry into a baryon asymmetry, linking neutrino physics to the existence of matter.

Type I: Collider Phenomenology – The "Invisible" Seesaw

- Because the right-handed neutrinos N_R are **gauge singlets**, they do not couple to the photon, W boson, or Z boson. Consequently, they cannot be directly produced via Drell–Yan processes at hadron colliders like the LHC.
- Moreover, if the seesaw scale M_R is near the GUT scale (10^{14} GeV), direct kinematic production is **completely impossible** – the center-of-mass energy of the LHC is only 14 TeV, far below 10^{14} GeV.
- Therefore, the only experimental probes of the minimal Type I seesaw are **indirect**:
  - **Neutrinoless double-beta decay** ($0\nu\beta\beta$), which tests the Majorana nature and the effective mass $m_{2\beta}$.
  - **Cosmological observations** of the sum of neutrino masses and the effective number of relativistic species.

Collider Invisibility

At its naturally high mass scale, Type I is essentially **invisible** to colliders.

Type II Seesaw: The Scalar Triplet

- We extend the SM by adding a heavy scalar triplet $\vec{\Delta} \sim (\mathbf{1}, \mathbf{3}, +2)$.
- To bridge the vector notation ($\vec{\Delta}$) with the physical fields, we expand it in the Pauli basis $\vec{\tau} = (\tau^1, \tau^2, \tau^3)$:

$$\mathbf{\Delta} \equiv \vec{\tau} \cdot \vec{\Delta} = \begin{pmatrix} \Delta^+ & \sqrt{2}\Delta^{++} \\ \sqrt{2}\Delta^0 & -\Delta^+ \end{pmatrix}$$

- This single object contains the three physical degrees of freedom:
  1. Δ^{++}: The doubly-charged scalar (the "smoking gun").
  2. Δ^+: The singly-charged scalar.
  3. Δ^0: The neutral scalar that acquires an induced VEV.
- **Mass Generation:** The neutral component Δ^0 acquires a tiny VEV (v_Δ), directly generating the Majorana mass matrix $m_\nu = Y_\Delta v_\Delta$.

Experimental Hook

Unlike the sterile Type I, the triplet's charges allow for direct Drell-Yan production and same-sign dilepton decays ($\Delta^{++} \to \ell^+\ell^+$).

Type II Seesaw: The Lagrangian and Lepton Number

- Using vector notation ($\vec{\tau}$ for Pauli matrices), the relevant additions to the Lagrangian are:

$$\mathcal{L}_{\rm II} = \mathcal{L}_{\rm SM} + |D_\mu \vec{\Delta}|^2 - M_\Delta^2 |\vec{\Delta}|^2 + \frac{1}{2}\left[Y_\Delta^{\alpha\beta} (L_\alpha \epsilon \vec{\tau} L_\beta) \cdot \vec{\Delta} + \mu (\phi \epsilon \vec{\tau} \phi) \cdot \vec{\Delta}^* + \text{h.c.} \right]$$

- $Y_\Delta^{\alpha\beta}$ is a **symmetric Yukawa coupling matrix** (coupling two identical fermions).
- The term with μ (mass dimension $+1$) is a **trilinear coupling** connecting the scalar triplet to two SM Higgs doublets.
- **Lepton number violation:** To conserve L in the Yukawa term, we must assign $L(\Delta) = -2$. Therefore, the trilinear μ term explicitly **violates lepton number by two units** ($\Delta L = 2$), the essential requirement for Majorana mass.

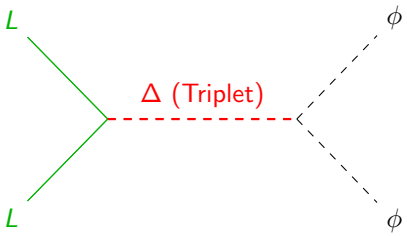
Type II: The Induced VEV and Light Mass

- After electroweak symmetry breaking ($\langle\phi\rangle = v/\sqrt{2}$), the trilinear μ term generates a linear "tadpole" in the scalar potential, forcing the neutral component of the triplet to acquire a **small, induced VEV**:
$$v_\Delta \equiv \langle\Delta^0\rangle \simeq \frac{\mu\, v^2}{2\, M_\Delta^2}$$
- Once the triplet acquires this VEV, the Yukawa term directly generates the **Majorana mass** for the active neutrinos:
$$\boxed{m_\nu = Y_\Delta v_\Delta \simeq Y_\Delta \frac{\mu\, v^2}{2 M_\Delta^2}}$$
- **Key difference from Type I:** Here m_ν is **linear** in the Yukawa coupling Y_Δ, not quadratic.

Mass Scale Implication

If $M_\Delta \sim 10^{14}$ GeV and $\mu \sim M_\Delta$, we get an unphysically small $m_\nu \sim 10^{-6}$ eV (assuming $Y_\Delta \sim 1$). To get the observed $m_\nu \sim 0.1$ eV, we naturally require a lower M_Δ (e.g., TeV scale), making Type II **highly testable at colliders**.

Type II: Tree-Level Feynman Diagram



- The diagram shows the **exchange of a heavy scalar triplet** Δ between two lepton doublets (green) and two Higgs doublets.
- The lepton-number-violating trilinear coupling μ connects the triplet to the Higgs VEVs, allowing the L number to flow through the diagram.
- After integrating out the heavy triplet, the effective Weinberg operator emerges.

Type II: Spectacular Collider Signatures

- Unlike the sterile singlets of Type I, the scalar triplet Δ **carries electroweak charges** and therefore couples to the W, Z, and γ bosons. This allows **direct production** at hadron colliders via Drell–Yan processes: $q\bar{q} \to \gamma^*/Z^* \to \Delta^{++}\Delta^{--}$.
- The most striking signature is the **doubly-charged scalar** Δ^{++}. It decays almost exclusively into a pair of **same-sign leptons**:

$$\Delta^{++} \to \ell_\alpha^+ \ell_\beta^+.$$

- This provides an **essentially background-free** signal at the LHC because same-sign dilepton events are extremely rare in the SM. The invariant mass distribution of the two leptons would show a sharp peak at $M_{\Delta^{++}}$.

Current and Future Limits

Current LHC searches have set lower limits $M_{\Delta^{++}} \gtrsim 800$ GeV (depending on the branching ratios). Future colliders like the HL-LHC and FCC could probe masses up to several TeV.

Type III Seesaw: The Fermion Triplet

- Introduced as the final tree-level UV completion, adding heavy fermions N_i that transform as an $SU(2)_L$ **triplet**: $(\mathbf{1}, \mathbf{3}, 0)$.
- Expanding the vector $\vec{N}$ identifies the three physical degrees of freedom:
  1. N^3: A neutral Majorana fermion (mixes with active neutrinos).
  2. $N^{\pm}$: Heavy charged fermions, defined as $(N^1 \mp iN^2)/\sqrt{2}$.
- Unlike the Type I singlet, these fields carry **weak isospin charges**, allowing them to be directly produced at the LHC via W and Z boson exchange.

Type III Seesaw: The Lagrangian

- Following the Strumia convention, the Lagrangian for the triplet N is:

$$\mathcal{L}_{\rm III} = \mathcal{L}_{\rm SM} + \bar{N}_i i \not{D} N_i - \left[\lambda^N_{ij} \vec{N}_i \cdot (\overline{L}_j \vec{\tau} \tilde{\phi}) + \frac{M^N_{ij}}{2} \vec{N}_i \cdot \vec{N}_j + \text{h.c.} \right]$$

- **Kinetic Term:** The covariant derivative $\not{D}$ couples the heavy fermions to SM gauge bosons ($W^\pm, Z$).
- **Yukawa Term (λ^N):** Couples the lepton doublet L to the triplet via the conjugate Higgs $\tilde{\phi} = \epsilon \phi^*$.
- **Majorana Term (M_N):** A bare mass matrix. Because the triplet has zero hypercharge, this term is **explicitly allowed** by gauge invariance.

Notation adapted from: Strumia & Vissani, hep-ph/0606054, Eq. 2.8

Type III: Mass Matrix and Seesaw Formula

- After EWSB ($\langle\phi\rangle = v/\sqrt{2}$), the neutral component N^3 mixes with the active neutrinos. The mass matrix matches the Type I structure:

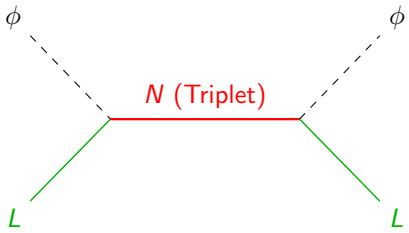
$$\mathcal{M} = \begin{pmatrix} 0 & m_D \\ m_D^T & M_N \end{pmatrix}, \quad \text{where } m_D = \lambda^N \frac{v}{\sqrt{2}}$$

- In the limit $M_N \gg m_D$, we recover the light neutrino mass matrix:

$$\boxed{m_\nu \simeq -\frac{v^2}{2}\lambda^N (M_N)^{-1} (\lambda^N)^T}$$

- **Signatures:** While the mass formula is identical to Type I, the existence of $N^\pm$ leads to spectacular **multilepton final states**.

Type III: Tree-Level Feynman Diagram



- The diagram is topologically identical to Type I, but the internal line represents the **heavy fermion triplet** N instead of a singlet N_R.
- Its triplet nature allows for the exchange of both neutral (N^3) and charged ($N^\pm$) states.
- This coupling to gauge bosons is the key differentiator for collider phenomenology, enabling direct production that is impossible in the minimal Type I model.

Type III: Collider Signatures and Current Limits

- Because the triplet carries weak isospin, the states couple to electroweak gauge bosons. The neutral N^3 and charged $N^\pm$ are produced via W/Z exchange: $q\bar{q} \to W^\pm \to N^\pm N^3$.
- The decay modes of these heavy fermions are exceptionally rich:

$$N^3 \to \nu Z,\ \ell^\pm W^\mp, \qquad N^\pm \to \ell^\pm Z,\ \nu W^\pm,\ \ell^\pm h.$$

- These processes lead to **multilepton final states** (e.g., three or four leptons) often accompanied by missing transverse energy.

LHC Limits

LHC searches for these "smoking gun" signatures have set lower limits on the triplet mass at approximately $M_N \gtrsim 800$–900 **GeV**. This makes Type III a **testable seesaw variant**: if the neutrino mass scale is linked to TeV-scale physics, these heavy leptons could be discovered at the HL-LHC.

Comparison of the Three Seesaw Mechanisms

	Type I	**Type II**	**Type III**
Mediator	$N_R : (\mathbf{1}, \mathbf{1}, 0)$	$\vec{\Delta} : (\mathbf{1}, \mathbf{3}, +2)$	$\vec{N} : (\mathbf{1}, \mathbf{3}, 0)$
Spin	Fermion (1/2)	Scalar (0)	Fermion (1/2)
L **Violation**	M_R (Majorana)	μ (Trilinear)	M_N (Majorana)
Mass Formula	$-m_D M_R^{-1} m_D^T$	$Y_\Delta \frac{\mu v^2}{2M_\Delta^2}$	$-m_D M_N^{-1} m_D^T$
m_ν **vs.** Y	Quadratic	Linear	Quadratic
EW Production	None (Singlet)	Yes (γ, W, Z)	Yes (W, Z)
Collider Sig.	Invisible	$\Delta^{++} \to \ell^+\ell^+$	Multilepton tracks

Unifying Feature

All three mechanisms naturally explain the smallness of neutrino masses via heavy-scale suppression ($1/M$) and can generate the observed baryon asymmetry of the universe through high-scale **Leptogenesis**.

Sterile Neutrinos: Gauge Singlets and the Mass Matrix

- **Sterile neutrinos** N_R are SM gauge singlets $(\mathbf{1}, \mathbf{1}, 0)$.
- They do *not* couple to W, Z, or γ – only via **mixing** with active neutrinos.
- The general Lagrangian (with $i = 1, \ldots, n$ sterile fields):

Sterile Sector Lagrangian

$$\mathcal{L} \supset i\overline{N_{Ri}}\partial\!\!\!/ N_{Ri} - \frac{1}{2}\left(\overline{N^c_{Ri}}\,(M_R)_{ij}\,N_{Rj} + \overline{\nu_{La}}\,(m_{ab})\,\nu^c_{Lb} + 2\,\overline{\nu_{La}}\,(m_D)_{ai}\,N_{Ri} + \text{h.c.}\right)$$

- In the $(\nu_L, N^c_R)^T$ basis, the full mass matrix is:

$$\mathcal{M} = \begin{pmatrix} m & m_D \\ m_D^T & M_R \end{pmatrix}$$

- **Type I seesaw** is the special case with $m = 0$ (no active Majorana mass), $m_D = Y_\nu v/\sqrt{2}$, and M_R the heavy Majorana mass.

Sterile Neutrinos: Helicity Suppression and Regimes

- Detection of the sterile component requires a **helicity flip** (amplitude $\propto m_\nu/E$).
- For MeV–GeV neutrinos, suppression $\sim 10^{-12}$ in probability $\Rightarrow$ sterile states are **nearly invisible** in oscillation experiments.

Three Regimes (depending on m, m_D, M)

- **Regime 1 (Decoupled):** $m_D \ll m, M_R$ – no mixing, no oscillation effects.
- **Regime 2 (Pseudo-Dirac):** $m_D \gg m, M_R$ – nearly degenerate pairs, maximal mixing; constrained to $m, M \lesssim 10^{-9}$ eV.
- **Regime 3 (Seesaw):** $m \ll m_D \ll M_R$ – light active masses $\sim m_D^2/M$, heavy states $\sim M_R$ (almost sterile).

- **Type I seesaw** corresponds to Regime 3 with $m = 0$.

Cosmology and Sterile Neutrinos

- Light sterile neutrinos (mass $\lesssim$ eV) behave as **extra radiation** if they are fully thermalised.
- They contribute to the effective number of relativistic species:

$$N_{\rm eff} = N_\nu^{\rm SM} + \Delta N_{\rm eff}, \quad N_\nu^{\rm SM} = 3.046$$

- **Planck 2018 + BBN** constrain $\Delta N_{\rm eff} < 0.3$ (95% CL) – ruling out a fully thermalised additional sterile neutrino.

Sterile Neutrinos as Dark Matter?

- A sterile neutrino with mass $\sim$ keV can be a **warm dark matter** candidate.
- Produced via active–sterile mixing (Dodelson–Widrow mechanism).
- Heavily constrained by X-ray searches (non-observation of decay line) and Lyman-α forest.

- We later show **cosmological constraints** on $\sum m_\nu$ applies to active neutrinos; sterile neutrinos are additionally bounded by $N_{\rm eff}$ and structure formation.

The Weinberg Operator: The Unifying Concept

- Regardless of which UV completion (Type I, II, or III) is realized in nature, the low-energy effective theory is always the same: the **dimension-5 Weinberg operator**.
- This operator is the **only** way to generate Majorana masses for active neutrinos up to dimension 5, given the SM gauge symmetry and particle content.
- Its observation, via **neutrinoless double-beta decay** ($0\nu\beta\beta$), would provide direct evidence for lepton number violation and the Majorana nature of neutrinos.
- The inferred scale $\Lambda \sim 3 \times 10^{14}$ GeV is tantalizingly close to the GUT scale, suggesting a deep connection between neutrino mass and grand unification.

The Next Frontier

The next generation of $0\nu\beta\beta$ experiments (LEGEND-1000, nEXO, CUPID) will probe the inverted hierarchy region and may discover the decay, thereby confirming the Weinberg operator and the seesaw paradigm.

The Cosmic Neutrino Background: Decoupling

- In the early universe, the energy density of a relativistic species at temperature T is:

$$\rho(T) = \frac{g}{(2\pi)^3} \int \frac{4\pi p^3}{e^{p/T} \mp 1} dp = \begin{cases} \frac{\pi^2}{30} g T^4 & \text{(Bosons)} \\ \frac{7}{8} \frac{\pi^2}{30} g T^4 & \text{(Fermions)} \end{cases}$$

- Neutrinos were kept in thermal equilibrium via weak interactions ($\Gamma_W \sim G_F^2 T^5$).
- When the universe expanded and cooled to $T_{\text{dec}} \sim 1$ MeV, the weak interaction rate dropped below the Hubble expansion rate ($\Gamma_W < H$).
- At this point, neutrinos **decoupled** from the primordial plasma and began to free-stream, forming the Cosmic Neutrino Background (CνB).

Entropy Conservation and the Temperature Ratio

- Shortly after neutrino decoupling, the temperature dropped below the electron mass ($m_e \approx 0.511$ MeV), triggering electron-positron annihilation ($e^+e^- \to \gamma\gamma$).
- This process dumped entropy into the photon bath. The total comoving entropy $S \propto g_{*S} T^3 a^3$ is conserved.
- **Before annihilation** ($T > m_e$): Photons, e^-, and e^+ are in equilibrium.

$$g_{*S}^{\text{before}} = g_\gamma + \frac{7}{8}(g_{e^-} + g_{e^+}) = 2 + \frac{7}{8}(2+2) = \frac{11}{2}$$

- **After annihilation** ($T < m_e$): Only photons remain. $g_{*S}^{\text{after}} = g_\gamma = 2$.
- Because neutrinos had already decoupled, they missed this heating. Equating entropy before and after yields the present-day temperature ratio:

$$T_\nu = \left(\frac{g_{*S}^{\text{after}}}{g_{*S}^{\text{before}}}\right)^{1/3} T_\gamma = \left(\frac{4}{11}\right)^{1/3} T_\gamma \approx 1.95 \text{ K} \approx 1.7 \times 10^{-4} \text{ eV}.$$

Present-Day Number and Energy Density

- We can now calculate the present-day number density of the CνB. Using Fermi-Dirac statistics for the neutrinos and Bose-Einstein for the CMB photons, the ratio of their number densities is:

$$\frac{n_\nu}{n_\gamma} = \left(\frac{3}{4}\right)\left(\frac{T_\nu}{T_\gamma}\right)^3 = \left(\frac{3}{4}\right)\left(\frac{4}{11}\right) = \frac{3}{11}$$

- Given the measured CMB photon density $n_\gamma \approx 411\ \mathrm{cm}^{-3}$, the number density **per neutrino flavor** today is:

$$n_{\nu,i} = \frac{3}{11} n_\gamma \approx 112\ \mathrm{cm}^{-3}.$$

- Because $T_\nu \approx 10^{-4}$ eV is extremely low compared to the measured mass splittings ($\sqrt{\Delta m_{31}^2} \approx 0.05$ eV), at least two mass eigenstates are **non-relativistic** today.
- Therefore, their present-day energy density acts as matter, not radiation:

$$\rho_\nu = \sum_i m_i n_{\nu,i}.$$

Neutrino Fraction of the Cosmic Energy Budget

- The critical density of the universe is $\rho_c = 3H_0^2/(8\pi G)$. The neutrino density parameter is:

$$\Omega_\nu = \frac{\rho_\nu}{\rho_c} = \frac{\sum m_\nu}{93.14 \text{ eV } h^2},$$

  where $h = H_0/(100 \text{ km/s/Mpc})$.
- We define the **neutrino fraction of the total matter density**:

$$f_\nu \equiv \frac{\Omega_\nu}{\Omega_m}.$$

- For $\sum m_\nu = 0.1$ eV and $\Omega_m h^2 \approx 0.14$, we get $f_\nu \approx 0.008$ – small, but cosmologically significant.

Neutrino Kinematics: Hot vs. Cold Relics

- **Cold Dark Matter (CDM):** Decouples when it is highly non-relativistic. It has negligible thermal velocity and is easily trapped by gravitational wells, forming the scaffolding for structure.
- **Neutrinos (Hot/Warm Relics):** Decouple at $T \sim 1$ MeV, while they are **highly relativistic** ($T \gg m_\nu$).
- Long after decoupling, they retain a massive thermal velocity dispersion.
- As the universe expands, this velocity simply redshifts away inversely with the scale factor a:

$$v_{\text{th}} \propto \frac{1}{a}$$

The Physics of Free-Streaming

- Because of their immense thermal velocity, neutrinos do not stay put. They can easily escape from small gravitational potential wells.
- This escape process is called **free-streaming**.
- Physically, free-streaming acts as a cosmic "eraser." It completely washes out any neutrino density perturbations on distance scales smaller than the distance the neutrinos can travel.
- Neutrinos can only cluster and fall into gravitational wells that are physically larger than this **free-streaming horizon**.

The Comoving Free-Streaming Wavenumber

- To quantify this scale mathematically, we use the comoving wavenumber (k).
- **Definition:** A "comoving" scale factors out the expansion of the universe, allowing us to track the same physical structure over cosmic time.
- The boundary of the washout effect is defined by the comoving free-streaming wavenumber:

Free-Streaming Scale

$$k_{\mathrm{fs}}(a) = \sqrt{\frac{3}{2}}\,\frac{aH(a)}{v_{\mathrm{th}}(a)}$$

The Cosmological Consequence

For small spatial scales (large wavenumbers where $k \gg k_{\mathrm{fs}}$), neutrinos travel too fast to be captured. Their density contrast vanishes: $\delta_\nu \to 0$.

Fluid Equations for Cold Dark Matter

- Deep in the matter-dominated era, CDM is modelled as a **pressureless perfect fluid** in the Newtonian gauge.
- The continuity and Euler equations give:

$$\dot{\delta}_c + \frac{1}{a}\vec{\nabla} \cdot \vec{v}_c = 0, \qquad \dot{\vec{v}}_c + H\vec{v}_c = -\frac{1}{a}\vec{\nabla}\Phi.$$

- Taking the time derivative of the continuity equation and substituting the divergence of the Euler equation yields the **master evolution equation**:

$$\ddot{\delta}_c + 2H\dot{\delta}_c = \frac{1}{a^2}\nabla^2\Phi.$$

- The Poisson equation couples the gravitational potential Φ to the **total matter perturbation** δ_m:

$$\frac{1}{a^2}\nabla^2\Phi = 4\pi G\bar{\rho}_m\delta_m.$$

Suppressed Growth of CDM on Small Scales

- The total matter perturbation is the density-weighted sum of CDM and neutrinos:

$$\delta_m = (1 - f_\nu)\delta_c + f_\nu \delta_\nu.$$

- On small scales ($k \gg k_{\rm fs}$), free-streaming erases neutrino clustering: $\delta_\nu \to 0$, so $\delta_m \approx (1 - f_\nu)\delta_c$.
- Substituting into the Poisson equation and then into the CDM evolution equation gives:

$$\ddot{\delta}_c + 2H\dot{\delta}_c = 4\pi G \bar{\rho}_m (1 - f_\nu)\delta_c.$$

- During matter domination: $a \propto t^{2/3}$, $H = 2/(3t)$, and $4\pi G\bar{\rho}_m = \frac{3}{2}H^2 = \frac{2}{3t^2}$. Thus:

$$\ddot{\delta}_c + \frac{4}{3t}\dot{\delta}_c - \frac{2}{3t^2}(1 - f_\nu)\delta_c = 0.$$

Solving the Growth Equation

- Try a power-law solution $\delta_c \propto t^\alpha$. Substituting yields the characteristic polynomial:

$$\alpha^2 + \frac{1}{3}\alpha - \frac{2}{3}(1 - f_\nu) = 0.$$

- The growing mode is:

$$\alpha_+ = \frac{-1/3 + \sqrt{1/9 + \frac{8}{3}(1 - f_\nu)}}{2} = \frac{-1 + \sqrt{25 - 24 f_\nu}}{6}.$$

- Expand for $f_\nu \ll 1$: $\sqrt{25 - 24 f_\nu} \approx 5\left(1 - \frac{12}{25} f_\nu\right) = 5 - \frac{12}{5} f_\nu$.
- Hence:

$$\alpha_+ \approx \frac{-1 + 5 - \frac{12}{5} f_\nu}{6} = \frac{2}{3} - \frac{2}{5} f_\nu.$$

- Since $t \propto a^{3/2}$, we convert to scale-factor growth:

$$\delta_c(k \gg k_{\rm fs}) \propto a^{1 - \frac{3}{5} f_\nu}.$$

- CDM grows **slower** than the standard $\delta_c \propto a$ because the gravitational source is reduced by $(1 - f_\nu)$.

From CDM Growth to Power Spectrum Suppression (I)

- The CDM power spectrum P_c is proportional to the square of the density contrast: $P_c(k) \propto |\delta_c(k)|^2$.
- We compare the power spectrum today ($a_0 = 1$) in a universe with neutrinos vs. one without, normalizing at the time suppression begins (a_i):

$$\frac{P_c^{(f_\nu)}}{P_c^{(0)}} = \left[\frac{\delta_c(a_0, f_\nu)}{\delta_c(a_0, 0)}\right]^2 \approx \left[\frac{a_0^{1-\frac{3}{5}f_\nu}}{a_0^1}\right]^2 \quad \text{(normalized at } a_i\text{)}$$

- Accounting for the growth starting at a_i, the ratio simplifies to:

$$\frac{P_c^{(f_\nu)}}{P_c^{(0)}} \approx \left(\frac{a_0}{a_i}\right)^{-\frac{6}{5}f_\nu} \approx 1 - \frac{6}{5} f_\nu \ln \frac{a_0}{a_i}$$

- This term represents the **cumulative dynamical suppression** of the CDM scaffolding itself.

From CDM Growth to Power Spectrum Suppression (II)

Why is $\ln(a_0/a_i) \approx 5$?

- The suppression begins when neutrinos become non-relativistic (a_{nr}) or at matter-radiation equality (a_{eq}), whichever is later.
- For $m_\nu \sim 0.1$ eV, neutrinos become non-relativistic at $z \approx 100$ ($a \approx 0.01$).
- The number of "e-folds" of growth since that time is $\ln(1/0.01) \approx 4.6 \approx 5$.
- For modes relevant to galaxy surveys ($k \sim 0.1\ h/\text{Mpc}$), this provides a consistent logarithmic factor.

- Because neutrinos do not cluster at these scales, the observable total matter perturbation is $\delta_m \approx (1 - f_\nu)\delta_c$:

$$P_m(k) \approx (1 - f_\nu)^2 P_c(k) \approx (1 - 2f_\nu) P_c(k)$$

- **Total Suppression:** Combining the density effect ($-2f_\nu$) and the slowed growth ($-\frac{6}{5} \cdot 5f_\nu$):

$$\frac{\Delta P_m}{P_m} \approx -2f_\nu - 6f_\nu = \boxed{-8f_\nu}$$

The Master Formula for Structure Suppression

- Substituting $f_\nu = \Omega_\nu/\Omega_m$ and $\Omega_\nu = \sum m_\nu/(93.14\ \text{eV}\ h^2)$ gives the **absolute master formula** used by cosmological surveys:

$$\boxed{\frac{\Delta P(k)}{P(k)} \approx -8\left(\frac{\sum m_\nu}{\Omega_m h^2 \cdot 93.14\ \text{eV}}\right).}$$

- This equation shows that the suppression of the matter power spectrum is **linear in the sum of neutrino masses**.
- A larger $\sum m_\nu$ leads to a stronger suppression, especially on small scales ($k \sim 0.1\ h/\text{Mpc}$).

Cosmological Bounds: From Planck to DESI

- Extracting $\sum m_\nu$ requires breaking parameter degeneracies, notably between the late-time clustering amplitude (σ_8), the matter density (Ω_m), and the Hubble constant (H_0).
- **The Methodological Synergy:**
  - **Planck (CMB Lensing):** Extracts the deflection potential power spectrum ($C_\ell^{\phi\phi}$). This is an **amplitude probe**: it measures the integrated gravitational lensing caused by intervening Large Scale Structure. Massive neutrinos free-stream and suppress this structure, thereby directly reducing the net lensing signal.
  - **DESI (BAO):** Measures the Baryon Acoustic Oscillation scale across cosmic time. This is a **geometric probe**: it acts as a standard ruler to map the background expansion history $H(z)$ and angular diameter distance $D_A(z)$.
- **The Evolution of Limits:**
  - **Planck 2018 + BAO:** $\sum m_\nu < 0.12$ eV (95% C.L.) [**Planck:2018**].
  - **DESI 2024 + Planck PR4:** Unprecedented precision in BAO measurements has dramatically tightened the upper bound [**DESI:2024**]:

$$\boxed{\sum m_\nu < 0.072 \text{ eV} \quad (95\% \text{ C.L.})}$$

Implications for the Neutrino Mass Hierarchy

- Terrestrial oscillation experiments fix the mass-squared differences:

$$\Delta m_{21}^2 = (7.5 \pm 0.1) \times 10^{-5}\ \text{eV}^2, \qquad |\Delta m_{31}^2| \approx 2.5 \times 10^{-3}\ \text{eV}^2.$$

- These values imply a **minimum possible sum** of neutrino masses:
  - **Normal ordering (NO):** $\sum m_\nu \approx 0.06$ eV.
  - **Inverted ordering (IO):** $\sum m_\nu \approx 0.10$ eV.
- The DESI+Planck bound of 0.072 eV **robustly excludes the entire inverted ordering parameter space**.

Conclusion

Unless there are unrecognized systematic errors or extensions to ΛCDM, cosmology currently provides the strongest evidence that nature has chosen the **normal mass hierarchy**.